

THE SPATIAL DISTRIBUTION OF MARKUPS

Preliminary and Incomplete

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Abstract

How much does the geographic segmentation of the US economy matter for competition and market power? We answer this question using a quantitative spatial model with multi-establishment firms, oligopolistic competition, and endogenously variable markups, calibrated to match US manufacturing data across 170 Economic Areas. Our benchmark model implies an aggregate markup of 1.26 with substantial variation across locations, from 1.22 in greater New York to 1.35 in Honolulu. The welfare costs of markups are large, 5.8% on average, and considerably higher than the 3.7% implied by an equivalent model without geography and spatial frictions. The welfare costs are also very unevenly distributed, ranging from about 1% or less in large central locations like New York to as much as 20% in Honolulu.

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1 Introduction

How much does the geographic segmentation of the US economy matter for competition and market power? Despite considerable improvements in transportation and the introduction of new technologies that make it easier for firms to sell in more locations, trade in goods across the US remains quite segmented with substantial home bias in shipment patterns even in the absence of formal trade barriers (Wolf, 2000; Hillberry and Hummels, 2008). Does this pattern simply reflect the concentration of production in a few highly productive locations? Or would more trade *within* the US increase competition, reduce markups, and increase aggregate productivity?

We answer these questions using a quantitative spatial model with oligopolistic competition and endogenously variable markups. Our model features many geographically segmented locations and heterogeneous firms that can, in general, source goods from multiple locations and sell in multiple locations. Firms are heterogeneous in their productivity and in the number and geographic location of their establishments. Taking wages in each location as given, each firm chooses an optimal, comprehensive production plan for their set of establishments that determines that firm’s effective marginal cost of producing for each possible destination market. Given these firm-and-destination-specific marginal costs, oligopolistic competition as in Atkeson and Burstein (2008) determines the markups each firm charges in each of its destination markets. Equilibrium wages in each location are determined by local labor market clearing conditions.

Most work on the macroeconomic consequences of markups (e.g., Baqaee and Farhi, 2020; De Loecker, Eeckhout and Mongey, 2021; Edmond, Midrigan and Xu, 2023) assumes frictionless trade within a country, and hence implicitly assumes that the spatial structure within a country is irrelevant for competition and market power. Similarly, most work on quantitative spatial economics (e.g., Allen and Arkolakis, 2014; Redding and Rossi-Hansberg, 2017) assumes either perfect competition or monopolistic competition with constant markups — and so, despite the rich geography of these models, implicitly assumes that the spatial economy does not interact with the determinants of competition and market power. By contrast, our model allows for extensive geographic variation in economic conditions within and across sectors and locations and determines firm-and-location-specific markups endogenously, in general equilibrium, jointly with other macro outcomes.

Our analysis focuses on manufactured goods, for which we can bring together detailed data on firms, establishments, and trade flows across locations. This is also a setting where goods vary enormously in tradeability, production is geographically concentrated, and firms range from single plants to large multi-establishment enterprises selling nationwide. We show that spatial frictions are quantitatively significant even in this setting.

We calibrate our model to match the operations of some 220,000 US manufacturing firms organized into 363 6-digit NAICS sectors, with an average of some 600 firms per sector.

We take our geographical locations to be 170 Economic Areas as constructed by the Bureau of Economic Analysis. While most firms are small and have only one establishment, larger firms tend to have multiple establishments and multiple-establishment firms produce in a broad range of locations. We ensure every firm in the model reproduces a real firm’s *geographic footprint* — we place each firm’s establishments exactly where they appear in the data. We choose the parameters of our model governing the distribution of productivity across firms and the correlation between a firm’s establishment count and its productivity to match key facts on national sales concentration and the share of employment accounted for by multi-establishment firms. We parameterize sector-specific iceberg trade costs so that the model exactly reproduces sector-specific gravity regressions using trade flows from the Commodity Flows Survey for 3-digit NAICS manufacturing sectors.

The fact that we match these sector-specific gravity effects is important. These gravity effects determine the quantitative significance of the spatial frictions in each sector, i.e., they determine which sectors produce goods that are intrinsically less tradeable and, within a given sector, which locations are more central and which are more remote. In other words, these frictions play a key role in determining the spatial distribution of production and sales. Intuitively, we find that production concentration is larger and more dispersed than sales concentration, with the gap being greatest in low-gravity sectors where goods are easily shipped to multiple destination markets.¹ Although our benchmark calibration lies much closer to free trade than to autarky, the remaining spatial frictions are still enough to matter substantially for competition.

Our main finding is that the spatial frictions in the US economy place substantial limits on competition, leading to a quantitatively significant amplification of the aggregate losses due to market power. In high-gravity sectors, local firms face less competition, leading to higher markups and higher productivity losses due to misallocation. We show that the *reduction in competition* provided by spatial frictions is equivalent to removing a randomly chosen 90% of firms from the economy. An otherwise equivalent model *without geography* — calibrated to match the same national concentration facts as our benchmark — has the same amount of competition, as measured by the aggregate markup and sales concentration, as our benchmark model *with geography* only after we remove 90% of its firms.

The intuition for this result is as follows. There are on average 600 firms in each 6-digit NAICS sector. In a standard calibration of a model without geography, this means that there are already enough large firms competing with each other that competitive outcomes are relatively insensitive to changes in the number of firms. We show that, without geography, this insensitivity is pervasive for a very wide range of the number of firms —

¹In our benchmark model the local sales Herfindahl-Hirschman Index (HHI) is, on average, about 0.15, higher than the national sales HHI of 0.10 but lower than the local production HHI of 0.36 reported by [Autor, Patterson and Van Reenen \(2023\)](#). This ordering of concentration is consistent with [Benkard, Yurukoglu and Zhang \(2026\)](#) who study concentration in finely disaggregated consumer survey data.

doubling or halving the number of firms has negligible effects on national concentration, on the aggregate markup, and on the productivity losses due to misallocation. It takes a drastic reduction in the number of firms, down to about 60 firms per 6-digit NAICS sector, for the model without geography to replicate the reduction in competition that is generated by the spatial frictions we estimate in US data. With these frictions, most sales in many locations are accounted for by just a few firms, especially in high-gravity sectors.

Our finding that spatial frictions substantially limit competition — and that reducing these frictions would deliver large pro-competitive gains — stands in sharp contrast to a class of benchmark models of endogenously variable markups, such as [Bernard, Eaton, Jensen and Kortum \(2003\)](#), [Atkeson and Burstein \(2008\)](#), [Edmond, Midrigan and Xu \(2015\)](#), and [Arkolakis, Costinot, Donaldson and Rodríguez-Clare \(2019\)](#), in which changes in trade costs or in the number of competing firms generate modest, if any, movements in markups when (i) firms are heterogeneous in productivity, and (ii) the number of competitors is large. Since the typical narrowly-defined US manufacturing industry has on average 600 firms, this class of models would predict that pro-competitive effects in the US should be small. By contrast, in our model the extent of spatial frictions determines which firms actually compete head-to-head in any given destination market. Spatial frictions can leave a location served by only a small number of *effective* competitors, even when the same industry has many firms nationally. Reducing trade costs therefore expands the set of effective competitors faced by dominant firms in each destination market, lowering markups and reducing the misallocation they generate.

Accounting for geography and spatial frictions leads to substantially higher welfare costs of markups. We measure these costs by asking how much the representative consumer in each location would gain from eliminating markups. We find that the average welfare cost is high, about 5.8% in consumption-equivalent terms, considerably higher than the 3.7% we find in the model without geography. The welfare costs are also very unevenly distributed, ranging from about 1% in greater New York to as much as 20% in Honolulu. Abstracting from geography and spatial frictions not only substantially understates the average welfare costs of markups, but also prevents the model from capturing the large variation in the burden of market power across locations.

We show that our model is also consistent with key trends in concentration. Our model generates endogenously diverging trends in national concentration and local concentration, as in [Rossi-Hansberg, Sarte and Trachter \(2020\)](#), when trade costs fall. Our model interprets these ‘diverging trends’ as *convergence* towards a better-integrated national market — as trade costs fall from their benchmark levels, national sales concentration rises while local sales concentration falls. In this sense, national sales concentration may rise even as markets become more competitive, markups fall, and aggregate productivity rises. Trends in national concentration may be a poor guide to trends in aggregate market power.

Finally, we ask to what extent worker mobility can mitigate the welfare costs of markups.

Our benchmark features workers that are immobile across locations. In an extension we consider a setup where workers have heterogeneous preferences for different location-specific amenities, pinning down labor supply to each location. We find that, when we parameterize the model to match the same initial allocation of labor across locations as in our benchmark, the welfare costs of markups are almost identical.

Intranational trade frictions. A large body of evidence documents that trade within the US is geographically segmented — see e.g., [Wolf \(2000\)](#), [Hillberry and Hummels \(2008\)](#), and [Martínez-San Román, Mateo-Mantecón and Sainz-González \(2017\)](#). We take this evidence as a starting point and ask what it implies for competition and market power. [Ramondo, Rodríguez-Clare and Saborío-Rodríguez \(2016\)](#) quantify the importance of intranational trade frictions in an [Eaton and Kortum \(2002\)](#) model with perfect competition.

Macroeconomics of market power. Our paper builds on the literature studying the macro consequences of market power, including [Baqaee and Farhi \(2020\)](#), [De Loecker, Eeckhout and Unger \(2020\)](#), [De Loecker, Eeckhout and Mongey \(2021\)](#), and [Edmond, Midrigan and Xu \(2023\)](#). These papers treat the domestic economy as a single spatially integrated whole. We show that abstracting from spatial frictions like this leads to a quantitatively significant understatement of the macro effects of market power.

Spatial economics. Our paper also builds on the recent literature on quantitative spatial economics, including [Allen and Arkolakis \(2014\)](#), [Caliendo and Parro \(2015\)](#), [Redding \(2016\)](#), [Redding and Rossi-Hansberg \(2017\)](#), and [Fajgelbaum, Morales, Serrato and Zidar \(2019\)](#). This literature has developed rich quantitative models for studying the spatial distribution of economic activity and the effects of changes in trade policy for trade flows and welfare across space, but typically assumes constant markups. We show that allowing markups to respond endogenously to the spatial economic structure has quantitatively important implications for the level and distribution of market power across locations.

Spatial misallocation. Our work is closely related to two recent papers on the spatial distribution of markups. Like us, [Asturias, García-Santana and Ramos \(2019\)](#) develop an [Atkeson and Burstein \(2008\)](#) model with many locations but unlike us they abstract from multi-location firms. Similarly, [Franco \(2023\)](#) studies spatial misallocation across cities in a model of monopolistic competition with [Kimball \(1995\)](#) demand. He emphasizes the sorting of firms across locations, which we abstract from, but does not consider the implications of multi-location firms for the spatial distribution of markups.

Trends in concentration. This paper contributes to the extensive literature on US trends in concentration since the early 1980s, following [Grullon, Larkin and Michaely \(2019\)](#), [Autor,](#)

Dorn, Katz, Patterson and Van Reenen (2020), Amiti and Heise (2021), Ganapati (2021), and many others. A central debate in this literature concerns the divergence between national and local trends: Rossi-Hansberg, Sarte and Trachter (2020), using NETS data, find that local concentration has been declining even as national concentration has risen, while subsequent work using Census data finds mixed or opposing results (e.g., Autor, Patterson and Van Reenen, 2023; Smith and Ocampo, 2024). As argued by Benkard, Yurukoglu and Zhang (2026), Census-based measures classify economic activity by *production* rather than by *consumption*, and consumer access to substitutes is what ultimately determines producer market power. Using finely disaggregated consumer survey data, Benkard et al. and Neiman and Vavra (2023) both find decreasing local sales concentration.

Multi-establishment production. This paper also contributes to the recent literature on the importance of multi-establishment firms, following Jia (2008), Holmes (2011), Basker, Klimek and Van (2012), Foster, Haltiwanger, Klimek, Krizan and Ohlmacher (2016), Cao, Hyatt, Mukoyama and Sager (2022), Hsieh and Rossi-Hansberg (2023), and many others. Most closely related is Oberfeld, Rossi-Hansberg, Sarte and Trachter (2024), who study the endogenous spatial distribution of production in a model with constant markups, whereas we take the geographic footprint of production as given and ask what it implies for the endogenous spatial distribution of markups.

2 Model

The economy consists of many heterogeneous locations. Across the economy there are many heterogeneous firms that, in general, can source goods from multiple locations and sell in multiple locations. Firms compete *oligopolistically* in their destination markets. The economy is *geographically segmented* in two ways: (i) labor is immobile across locations,² with location-specific wages pinned down by local labor market clearing conditions, and (ii) goods shipments are subject to iceberg trade costs.

2.1 Environment

There are J locations indexed by $j, k = 1, \dots, J$. There is a continuum of sectors indexed by $s \in [0, 1]$. Within each sector there is a finite $N(s)$ firms indexed by $i = 1, \dots, N(s)$ that compete oligopolistically in their destination markets. Trade in goods is subject to sector-specific iceberg trade costs $\tau_{jk}(s) \geq 1$ with $\tau_{jj}(s) = 1$. A notational convention that we maintain throughout is that location j refers to the *source* of a good and location k refers to a *destination* so that $\tau_{jk}(s)$, say, refers to the sector-specific cost of shipping from j to k .

²We discuss an extension with *labor mobility* across locations in Section 7 below.

Location-specific final good. In each destination market k there is a non-tradeable final good produced under perfectly competitive conditions. This location-specific final good is given by a CES aggregate *across sectors*

$$C_k = \left(\int_0^1 C_k(s)^{\frac{\theta-1}{\theta}} ds \right)^{\frac{\theta}{\theta-1}}, \quad \theta > 1 \quad (1)$$

Then *within sectors*, output is given by a CES aggregate across the $N(s)$ firms in sector s

$$C_k(s) = \left(\sum_{i=1}^{N(s)} C_{ik}(s)^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}}, \quad \gamma > \theta \quad (2)$$

We assume $\gamma > \theta$ so that goods are more substitutable within sectors than across sectors.

Firms source from establishments in multiple locations. Each firm i selling in destination market k sources goods from establishments in multiple locations $j = 1, \dots, J$. Specifically, we assume that each firm i supplies k with a CES aggregate *across establishments*

$$C_{ik}(s) = \left(\sum_{j=1}^J c_{ijk}(s)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \quad \eta \geq \gamma \quad (3)$$

Two special cases are worth noting: (i) the case $\eta = \gamma$ where goods are equally substitutable within and across firms (within a given sector), and (ii) the case $\eta = +\infty$ where a given firm will source all of its output from its least-cost establishment.

Location-specific representative consumer. Each location j is populated by L_j identical workers each endowed with E_j efficiency units of labor. Labor is *immobile* across locations. Each worker inelastically supplies their E_j units of labor to the local labor market and receives location-specific wage W_j per efficiency unit. Aggregating the budget constraints of workers in location j gives

$$P_j C_j = W_j E_j L_j + \Pi_j \quad (4)$$

where Π_j denotes aggregate profits paid out to workers in location j .

Distribution of profits. We assume that firm ownership is perfectly diversified across locations with profits paid out in proportion to labor income

$$\Pi_j = \bar{\pi} W_j E_j L_j \quad (5)$$

This implies that every location has the same labor and profit income shares

$$\left(\frac{1}{1 + \bar{\pi}}, \frac{\bar{\pi}}{1 + \bar{\pi}} \right) \quad (6)$$

for some constant $\bar{\pi} \geq 0$ to be determined in equilibrium.

Demand system. This nested-CES setup implies that the demand for goods sourced from location j to be sold at destination k by firm i in sector s is given by

$$c_{ijk}(s) = \left(\frac{\tau_{jk}(s)p_{ijk}(s)}{P_{ik}(s)} \right)^{-\eta} \underbrace{\left(\frac{P_{ik}(s)}{P_k(s)} \right)^{-\gamma} \left(\frac{P_k(s)}{P_k} \right)^{-\theta}}_{=C_{ik}(s)} C_k \quad (7)$$

As usual, the location-specific final good and sector-level price indexes are given by

$$P_k = \left(\int_0^1 P_k(s)^{1-\theta} ds \right)^{\frac{1}{1-\theta}}, \quad P_k(s) = \left(\sum_{i=1}^{N(s)} P_{ik}(s)^{1-\gamma} \right)^{\frac{1}{1-\gamma}} \quad (8)$$

But in this setup there is now also an index for aggregating prices across establishments within each firm. This firm-level composite price is given by

$$P_{ik}(s) = \left(\sum_{j=1}^J (\tau_{jk}(s)p_{ijk}(s))^{1-\eta} \right)^{\frac{1}{1-\eta}} \quad (9)$$

We implicitly set $p_{ijk}(s) = +\infty$ for any firm i that does not sell in location k .

Production. Firm i in sector s is endowed with productivity $z_{ij}(s) \geq 0$ for goods produced at location j . For simplicity we assume that the output shipped by firm i from source j to destination k is linear in labor

$$y_{ijk}(s) = z_{ij}(s)l_{ijk}(s) \quad (10)$$

Resource constraints. Given the sector-specific iceberg trade costs $\tau_{jk}(s) \geq 1$, the resource constraints on the flow of output from j to k are simply

$$y_{ijk}(s) = \tau_{jk}(s)c_{ijk}(s) \quad (11)$$

Marginal cost. Taking the wage rate W_j as given, firm i can source goods from j for any destination k at marginal cost

$$\frac{W_j}{z_{ij}(s)} \quad (12)$$

Profits. Since a firm can supply destination k with goods sourced from establishments at any location j , the firm's profits from sales at k are given by

$$\Pi_{ik}(s) = \sum_{j=1}^J \left(p_{ijk}(s) - \frac{W_j}{z_{ij}(s)} \right) y_{ijk}(s) \quad (13)$$

A firm's total profits are then given by $\Pi_i(s) = \sum_{k=1}^J \Pi_{ik}(s)$. This objective is separable across destinations k and hence the firm maximizes total profits by maximizing profits in each destination k separately.

We characterize the firm's profit maximizing strategy in each destination k in two steps: (i) taking as given the firm's composite price for its destination market, $P_{ik}(s)$, we determine the least-cost way of servicing that destination with one unit of the firm's composite good, $C_{ik}(s) = 1$, then (ii) we characterize how the firm's price $P_{ik}(s)$ is determined through oligopolistic competition with the other firms servicing destination k . The first step implicitly gives us a characterization of the allocation of production across locations within a given firm. Given the first step, the second step is a nested-CES oligopoly problem familiar from [Atkeson and Burstein \(2008\)](#) and [Edmond, Midrigan and Xu \(2015\)](#).

Within-firm allocation. Taking as given $P_{ik}(s)$ and $C_{ik}(s) = 1$, for step (i) firm i chooses prices $p_{ijk}(s)$ for $j = 1, \dots, J$ to minimize the total cost of servicing destination k

$$\sum_{j=1}^J \frac{W_j}{z_{ij}(s)} y_{ijk}(s) = \sum_{j=1}^J \frac{\tau_{jk}(s) W_j}{z_{ij}(s)} \left(\frac{\tau_{jk}(s) p_{ijk}(s)}{P_{ik}(s)} \right)^{-\eta} \underbrace{C_{ik}(s)}_{=1} \quad (14)$$

subject to the firm-level price index (9). The Lagrangian for this problem can be written

$$\mathcal{L} = \sum_{j=1}^J \frac{\tau_{jk}(s) W_j}{z_{ij}(s)} \left(\frac{\tau_{jk}(s) p_{ijk}(s)}{P_{ik}(s)} \right)^{-\eta} - \lambda_{ik}(s) \sum_{j=1}^J \left(\left(\frac{\tau_{jk}(s) p_{ijk}(s)}{P_{ik}(s)} \right)^{1-\eta} - 1 \right) \quad (15)$$

where $\lambda_{ik}(s) \geq 0$ denotes the multiplier on the firm's constraint. The first order conditions for interior solutions simplify to

$$\eta \frac{\tau_{jk}(s) W_j}{z_{ij}(s)} = (\eta - 1) \lambda_{ik}(s) \left(\frac{\tau_{jk}(s) p_{ijk}(s)}{P_{ik}(s)} \right) \quad (16)$$

Rearranging this we see that, at the optimum, source prices satisfy

$$p_{ijk}(s) = \mu_{ik}(s) \frac{W_j}{z_{ij}(s)}, \quad \mu_{ik}(s) = \frac{\eta}{\eta - 1} \left(\frac{P_{ik}(s)}{\lambda_{ik}(s)} \right) \quad (17)$$

Hence the least-cost way to service destination k is to set a *destination-specific* markup $\mu_{ik}(s)$ that applies uniformly regardless of the source location j . The firm ‘prices to market’ in a way that reflects the demand and competitive conditions specific to market k . But by making the markup independent of j the firm *avoids distorting allocations within the firm*.

Plugging this expression for the source prices $p_{ijk}(s)$ back into the firm-level price index (9) and eliminating the multiplier gives

$$P_{ik}(s) = \mu_{ik}(s) \cdot \text{MC}_{ik}(s) \quad (18)$$

where

$$\text{MC}_{ik}(s) = \left(\sum_{j=1}^J \left(\frac{\tau_{jk}(s) W_j}{z_{ij}(s)} \right)^{1-\eta} \right)^{\frac{1}{1-\eta}} \quad (19)$$

denotes the firm’s marginal cost of servicing destination k with one unit of the composite good $C_{ik}(s)$. With this characterization of the within-firm allocation in hand, we can now turn to the strategic interactions between firms in each destination k .

Oligopolistic competition. For step (ii) we then need to characterize how the firm’s price $P_{ik}(s)$ is determined through oligopolistic competition with the other firms servicing destination k . Given the within-firm allocation we can use (7), (13) and (18) to write the firm’s profits from destination k

$$\begin{aligned} \Pi_{ik}(s) &= (P_{ik}(s) - \text{MC}_{ik}(s)) C_{ik}(s) \\ &= (P_{ik}(s) - \text{MC}_{ik}(s)) \left(\frac{P_{ik}(s)}{P_k(s)} \right)^{-\gamma} \left(\frac{P_k(s)}{P_k} \right)^{-\theta} C_k \end{aligned} \quad (20)$$

with each firm internalizing the effect of their price $P_{ik}(s)$ on the sector-level price index $P_k(s)$ in (8). Given our characterization of the within-firm allocation in the first step, this second step is a standard nested-CES oligopoly problem familiar from [Atkeson and Burstein \(2008\)](#) and [Edmond, Midrigan and Xu \(2015\)](#).

As is well known, this implies that each firm sets a markup of the form

$$\mu_{ik}(s) = \frac{\varepsilon_{ik}(s)}{\varepsilon_{ik}(s) - 1} \quad (21)$$

where the demand elasticity $\varepsilon_{ik}(s)$ facing firm i is endogenous to the firm’s sales share in destination k . For our benchmark model we assume that each destination market is characterized by *Cournot competition*. With this specification, the demand elasticity works out to be a sales-weighted harmonic average of the elasticities of substitution within and

across sectors

$$\varepsilon_{ik}(s) = \left(\omega_{ik}(s) \frac{1}{\theta} + (1 - \omega_{ik}(s)) \frac{1}{\gamma} \right)^{-1} \quad (22)$$

where $\omega_{ik}(s)$ denotes the market share of firm i in destination market k

$$\omega_{ik}(s) := \frac{P_{ik}(s)C_{ik}(s)}{\sum_{i=1}^{N(s)} P_{ik}(s)C_{ik}(s)} = \frac{P_{ik}(s)^{1-\gamma}}{\sum_{i=1}^{N(s)} P_{ik}(s)^{1-\gamma}} \quad (23)$$

Since the elasticity of substitution across firms is larger than across sectors, $\gamma > \theta$, the demand elasticity $\varepsilon_{ik}(s)$ facing a firm is lower for firms with larger market shares in destination k . Firms that are small within a given market are mostly competing with other firms within the same sector and so face a relatively high demand elasticity, approaching the within-sector elasticity γ as $\omega_{ik}(s) \rightarrow 0$. At the other extreme, firms that are large within a given market are mostly competing with firms in other sectors and so face a relatively low demand elasticity, approaching the across-sector elasticity θ as $\omega_{ik}(s) \rightarrow 1$.

While intuitive, this discussion is incomplete. It simply takes market shares $\omega_{ik}(s)$ as exogenous and traces out the implications of those market shares for markups $\mu_{ik}(s)$. But in this model, markups and market shares are *jointly determined* as part of a larger fixed-point problem. To solve this problem, it turns out to be convenient to first combine (21) and (22) to write the inverse markup as a linear function of the sales share

$$\frac{1}{\mu_{ik}(s)} = 1 - \frac{1}{\varepsilon_{ik}(s)} = \frac{\gamma - 1}{\gamma} - \left(\frac{1}{\theta} - \frac{1}{\gamma} \right) \omega_{ik}(s) \quad (24)$$

From which we see that indeed a firm's markup is *strictly increasing* in its market share.

To obtain the second condition we need, we substitute prices $P_{ik}(s) = \mu_{ik}(s)\mathbf{MC}_{ik}(s)$ into (23) to get

$$\omega_{ik}(s) = \frac{(\mu_{ik}(s)\mathbf{MC}_{ik}(s))^{1-\gamma}}{\sum_{i=1}^{N(s)} (\mu_{ik}(s)\mathbf{MC}_{ik}(s))^{1-\gamma}} \quad (25)$$

Here we see that, conditional on other firms' markups, each firm's market share is *strictly decreasing* in its markup. Together, equations (24) and (25) are two equations in two unknowns that jointly determine the markups $\mu_{ik}(s)$ and market shares $\omega_{ik}(s)$ for each i, k and s . Notice that the interactions between firms within a given market enter only through the denominator in (25) and that market shares are homogenous of degree zero in the markups.

Eliminating the market shares between these we have a single fixed point condition

$$\frac{1}{\mu_{ik}(s)} = \frac{\gamma - 1}{\gamma} - \left(\frac{1}{\theta} - \frac{1}{\gamma} \right) \frac{(\mu_{ik}(s)\mathbf{MC}_{ik}(s))^{1-\gamma}}{\sum_{i=1}^{N(s)} (\mu_{ik}(s)\mathbf{MC}_{ik}(s))^{1-\gamma}} \quad (26)$$

This condition implicitly determines the distribution of markups $\mu_{ik}(s)$ within and across locations as a function of the distribution of marginal costs $\text{MC}_{ik}(s)$ within and across locations. The marginal costs $\text{MC}_{ik}(s)$ are exogenous to each firm but, because they depend on the wages W_j , still need to be determined in equilibrium.

General equilibrium. The equilibrium of the model is pinned down by labor market clearing in each local labor market. The labor market in each location j clears when the total supply of efficiency units of labor $E_j L_j$ equals the total labor demand in that location

$$E_j L_j = \int_0^1 \sum_{k=1}^J \sum_{i=1}^{N(s)} l_{ijk}(s) ds \quad (27)$$

Solving the model. We solve the model by Newton methods on a high-dimensional system of nonlinear equations. We initialize the algorithm using the solution of an alternative model which is identical except that profits are distributed uniformly across locations rather than proportionately. This alternative model can be solved much more efficiently and provides a good initial guess for the Newton algorithm we use to solve the benchmark model. The full details of our computational procedure are given in the Appendix.

We next briefly outline how the spatial distribution of markups $\mu_{ik}(s)$ affects aggregate productivity within and across locations.

2.2 Aggregation

Underlying all our aggregation results is an endogenous bilateral *productivity network*, a collection of productivity levels $\bar{z}_{jk}(s)$ for each sector s that forms a graph on the nodes j, k with directed edges from origins j to destinations k . The results below are obtained for the benchmark case $\eta = \gamma$, where goods are equally substitutable within and across firms. For this case we can give an exact closed-form decomposition of sector-level productivity $Z_k(s)$ at each destination k in terms of this underlying bilateral productivity network $\bar{z}_{jk}(s)$.³

Productivity network. To derive this productivity network, first let $\bar{c}_{jk}(s)$ denote the composite formed from goods shipped from j to k within a given sector

$$\bar{c}_{jk}(s) = \left(\sum_{i=1}^{N(s)} c_{ijk}(s)^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}} \quad (28)$$

³More generally, with $\eta > \gamma$, our expressions for firm-level markups (no distortions within the firm) and sector-level markup aggregates go through, but the bilateral decomposition is not available in closed form.

To write the bilateral composite $\bar{c}_{jk}(s)$ this way we make use of the fact that with $\eta = \gamma$ the quantity $c_{ijk}(s)$ is proportional to $p_{ijk}(s)^{-\gamma}$ across firms i for any source-destination pair j, k . Now let $\bar{l}_{jk}(s) = \sum_i l_{ijk}(s)$ denote the labor used to produce this composite and let $\bar{y}_{jk}(s) = \tau_{jk}(s)\bar{c}_{jk}(s)$ denote the amount of this composite that has to be produced at j for $\bar{c}_{jk}(s)$ to arrive at k . The productivity network is then given by the collection of $\bar{z}_{jk}(s) := \bar{y}_{jk}(s)/\bar{l}_{jk}(s)$. In keeping with this notation, let $\bar{\mu}_{jk}(s)$ denote the implied markup, satisfying $\bar{p}_{jk}(s) = \bar{\mu}_{jk}(s)W_j/\bar{z}_{jk}(s)$ where

$$\bar{p}_{jk}(s) = \left(\sum_{i=1}^{N(s)} p_{ijk}(s)^{1-\gamma} \right)^{\frac{1}{1-\gamma}} = \left(\sum_{i=1}^{N(s)} \left(\frac{\mu_{ik}(s)}{z_{ij}(s)} \right)^{1-\gamma} \right)^{\frac{1}{1-\gamma}} W_j \quad (29)$$

is the price index for the composite good. As in [Edmond, Midrigan and Xu \(2015, 2023\)](#), we then obtain

$$\bar{z}_{jk}(s) = \left(\sum_{i=1}^{N(s)} \left(\frac{\mu_{ik}(s)}{\bar{\mu}_{jk}(s)} \right)^{-\gamma} z_{ij}(s)^{\gamma-1} \right)^{\frac{1}{\gamma-1}} \quad (30)$$

Notice that in the special case of no dispersion in markups this reduces to a standard CES productivity index that depends only on the exogenous firm-level productivity $z_{ij}(s)$. More generally, markup dispersion reduces $\bar{z}_{jk}(s)$ below this benchmark. Notice also that $\bar{z}_{jk}(s)$ does not depend on the trade costs $\tau_{jk}(s)$. This is because, within a given sector s , the trade costs $\tau_{jk}(s)$ apply to all firms shipping from j to k equally.

With this expression for $\bar{z}_{jk}(s)$ in hand, the markup $\bar{\mu}_{jk}(s)$ on the composite good is

$$\bar{\mu}_{jk}(s) = \left(\sum_{i=1}^{N(s)} \frac{1}{\mu_{ik}(s)} \omega_{ijk}(s) \right)^{-1} = \frac{\sum_{i=1}^{N(s)} \mu_{ik}(s)^{1-\gamma} z_{ij}(s)^{\gamma-1}}{\sum_{i=1}^{N(s)} \mu_{ik}(s)^{-\gamma} z_{ij}(s)^{\gamma-1}} \quad (31)$$

where

$$\omega_{ijk}(s) = \left(\frac{p_{ijk}(s)}{\bar{p}_{jk}(s)} \right)^{1-\gamma} = \left(\frac{\mu_{ik}(s)}{z_{ij}(s)} \frac{\bar{z}_{jk}(s)}{\bar{\mu}_{jk}(s)} \right)^{1-\gamma} \quad (32)$$

denotes the sales share of firm i in shipments from j to k . In short, as in [Edmond, Midrigan and Xu \(2015, 2023\)](#), the markup on the composite good is a sales-weighted harmonic average of the firm-level markups $\mu_{ik}(s)$ for destination k .

Location-specific productivity and markups. With the bilateral productivity network $\bar{z}_{jk}(s)$ and associated markups $\bar{\mu}_{jk}(s)$ determined, we can aggregate further to get measures of location-specific productivity and markups. To this end, let $\bar{L}_k(s) := \sum_j \bar{l}_{jk}(s)$ denote the labor used *across all source locations* j to produce for destination k . Then let $\bar{Z}_k(s) := C_k(s)/\bar{L}_k(s)$ denote the real consumption at destination k per unit of this total labor input.

Using $\eta = \gamma$, we can write $C_k(s)$ as an aggregate of the bilateral composite $\bar{c}_{jk}(s)$ and then express the productivity index $\bar{Z}_k(s)$ in terms of the bilateral productivity network $\bar{z}_{jk}(s)$ and markups $\bar{\mu}_{jk}(s)$, giving

$$\bar{Z}_k(s) = \left(\sum_{j=1}^J \left(\frac{\bar{\mu}_{jk}(s)}{\bar{\mu}_k(s)} \right)^{-\gamma} \left(\frac{\bar{z}_{jk}(s)}{\tau_{jk}(s)} \right)^{\gamma-1} \left(\frac{W_j}{\bar{W}_k(s)} \right)^{-\gamma} \right)^{\frac{1}{\gamma-1}} \quad (33)$$

where $\bar{\mu}_k(s)$ denotes the location-specific markup, which again can be written as sales-weighted harmonic average of the underlying $\bar{\mu}_{jk}(s)$, and where $\bar{W}_k(s) = \sum_j W_j \bar{l}_{jk}(s) / \bar{L}_k(s)$ is the weighted average of source wages. This expression for $\bar{Z}_k(s)$ is similar to that given for the productivity nodes $\bar{z}_{jk}(s)$ in equation (30) above, but differs in two ways. First, the expression for $\bar{Z}_k(s)$ also depends on trade costs $\tau_{jk}(s)$, since trade costs reduce the contributions that high-productivity sources j make to destinations k that are costly to ship to. Second, the expression for $\bar{Z}_k(s)$ also depends on the relative wage $W_j / \bar{W}_k(s)$, which is absent from the expression for $\bar{z}_{jk}(s)$ since those productivity nodes refer to firms who are all paying the same wage W_j to produce in j .

Overall we see that markup dispersion reduces aggregate productivity, both because markup dispersion *across firms* within a given destination k directly reduces productivity $\bar{z}_{jk}(s)$ at each node in the productivity network, as in equation (30), and because for any given network of $\bar{z}_{jk}(s)$, markup dispersion *across locations*, reduces aggregate productivity $\bar{Z}_k(s)$ at each destination k , as in equation (33). In this sense, the spatial dispersion in markups creates *endogenous misallocation*, both across firms and across locations.

3 Quantifying the model

In this section we outline our benchmark parameterization and calibration strategy and present our model's implications for national and local sales concentration.

3.1 Benchmark parameterization

Our geographical locations are Economic Areas (EAs) constructed by the US Bureau of Economic Analysis. There are $J = 170$ EAs in our data. Each EA is built around an urban core, either larger Metropolitan Statistical Areas (MSAs) or smaller Micropolitan Statistical Areas, along with adjacent counties with strong commuting ties. Importantly, these EAs are built to reflect the fact that economic activity spans administrative/jurisdictional boundaries. For example, the Chicago EA includes both the Chicago metropolitan area along with other counties in Illinois, Indiana, and Wisconsin where workers have strong connections to Chicago. EAs are very diverse in size, ranging from the greater Los Angeles

area which accounts for around 14.6% of total employment, all the way down to Scottsbluff, Nebraska (near the Wyoming border) which accounts for 0.0012% of total employment:

1	Los Angeles-Riverside-Orange County	14.6%	of employment
2	New York-North New Jersey-Long Island	7.2%	
3	Chicago-Gary-Kenosha	6.9%	
	:		
169	San Angelo, TX	0.0013%	
170	Scottsbluff, NE-WY	0.0012%	

Sectors. We calibrate our model to match the operations of firms in 363 NAICS 6-digit manufacturing sectors, examples of which include *breakfast cereal* (sector 311230), *ready-mix concrete* (327320), *aircraft engine & engine parts* (336412), *optical instrument & lens manufacturing* (333314), and *wood kitchen cabinet & countertop manufacturing* (337110).

Labor supply. For each $j = 1, \dots, J$ we measure the number of workers L_j as manufacturing employment from the County Business Patterns (CBP) aggregated to the EA level. We likewise measure the wage bill from the CBP aggregated to the EA level and choose the efficiency units E_j for each location so that the wage bill in our model for each location $W_j E_j L_j$ matches the wage bill measured in the data for that location.

Firms and establishments. On average there are about 600 firms nationally per NAICS 6-digit sector, most of which are small. But there is considerable variation in the number of firms across sectors, ranging from a low of $N(s) = 8$ firms nationally for *custom roll forming* (sector 332114), a specialized manufacturing activity focused on the metal forming process, to $N(s) = 11,492$ for *machine shops* (332710) and $N(s) = 14,279$ for *commercial printing* (323111), which covers printing of stationery, advertising materials, etc. All together we have $N = \sum_s N(s) = 219,365$ firms. We set the number of establishments for each firm by counting the number of EA locations where firm i has at least one establishment in National Establishment Time Series (NETS) county-level data aggregated to the EA level. So, for example, if a given firm has an establishment in each of two counties that belong to the same EA we call that ‘one’ establishment for the purposes of our model.

Firms and establishment locations. We populate the economy by placing each firm’s establishments exactly where they appear in the data, so every model firm mirrors a real firm’s geographic footprint. Let $\mathcal{J}_i(s) \subseteq \{1, \dots, J\}$ denote the set of locations where firm i has establishments and let $n_i(s) = \sum_j \mathbf{1}\{j \in \mathcal{J}_i(s)\}$ denote the number of locations where firm i has establishments. We take this set $\mathcal{J}_i(s)$ exactly from the data, as [Figure 1](#) illustrates for two sectors, *semi-conductors & electronic components* and *ready-mix concrete*.

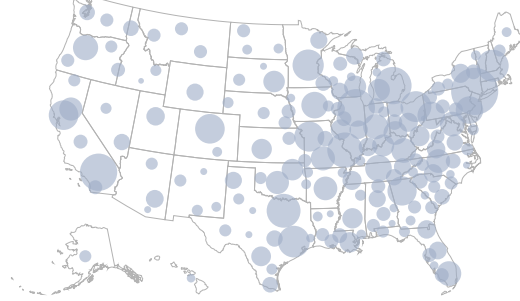
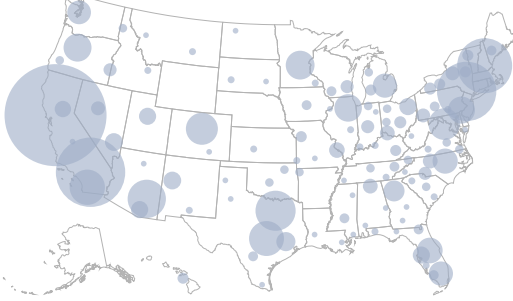
Figure 1: Exact geographic footprints of firms and sectors

A. *Semiconductors & Electronic Components*

B. *Ready-Mix Concrete*

Sector-level: 1471 firms, 1843 establishments

Sector-level: 1915 firms, 2162 establishments



Example: one firm with 44 establishments

Example: one firm with 74 establishments



We match the exact patterns of geographic dispersion in locations at the NAICS 6-digit sector, firm- and establishment-levels. The top row illustrates the sector-level dispersion for *semiconductors & electronic components* (sector 334413) and *ready-mix concrete* (327320). In the top row, each dot is an Economic Area (EA), with area proportional to the number of establishments the sector operates there. The bottom row chooses a real firm from each sector to illustrate the firm- and establishment-level footprints that we match. Each firm's establishments are placed at the EA locations $\mathcal{J}_i(s)$ where the firm operates in the data and joined by a minimum spanning tree.

Firm-level productivity fixed effects. We abstract from location-specific firm productivity effects and assume that firm-level productivity can be written

$$z_{ij}(s) = z_i(s) \cdot \mathbf{1}\{j \in \mathcal{J}_i(s)\} \quad (34)$$

In other words, a firm's productivity is equal to the firm-level productivity fixed effect $z_i(s)$ in all locations where it has an establishment and zero in all locations where it has no establishments. Since we have taken the set of locations $\mathcal{J}_i(s)$ for each firm directly from the data, the only thing left to do is to assign these firm-level productivity fixed effects.

To assign the firm-level productivity fixed effects, we simulate a large number of paired uniform ranks (u, v) using a Gumbel copula

$$\mathcal{C}(u, v) = \exp \left(- \left[(-\ln u)^{\frac{1}{1-\rho}} + (-\ln v)^{\frac{1}{1-\rho}} \right]^{1-\rho} \right) = \text{Gumbel Copula}(\rho) \quad (35)$$

We then transform the first rank into a Pareto productivity draw $z = F^{-1}(u)$ where

$$F(z) = 1 - z^{-\xi} = \text{Pareto}(\xi)$$

and transform the second rank into an empirical draw of establishment counts. For each establishment-count category, e.g., all firms with n establishments, we take the simulated pool of productivity-count pairs and condition on n to obtain candidate productivity draws. We then randomly sample the exact number of real firms in that category and assign these productivity levels to the corresponding firms in their real geographic locations.

EXAMPLE. Suppose in the data there are 1,123 firms that have $n = 10$ establishments. In the model, we can place the establishments of each of these 1,123 firms into the exact EA that they are in the data. But what we do not observe is their firm-level productivity. From the Gumbel simulation, however, we have an extremely large set of {productivity, establishment-count} pairs. We drop all pairs whose establishment count is not 10, which leaves us with a still very large sample of productivity draws from the distribution of firm-level productivity conditional on $n = 10$. We then randomly sample 1,123 productivity draws with replacement from this conditional distribution and assign them to the 1,123 firms we have already placed in their real geographic locations.

With this procedure in place, we are left with two parameters to pin down, the the Pareto tail ξ and the Gumbel rank correlation parameter ρ , as discussed further below.⁴

Trade costs. Following [Caliendo, Parro, Rossi-Hansberg and Sarte \(2018\)](#), we parameterize the sector-specific iceberg trade costs $\tau_{jk}(s)$ by assuming a sector-specific log-linear relationship between trade costs and physical distance d_{jk}

$$\ln \tau_{jk}(s) = \delta(s) \ln d_{jk} \tag{36}$$

This gives us a further set of parameters to pin down, the trade cost coefficients $\delta(s)$.

3.2 Calibration

Calibration strategy. We assign values to several conventional parameters that are held constant throughout all our quantitative exercises. We calibrate the remaining parameters internally using the simulated method of moments. We calibrate the parameters governing

⁴For the Gumbel copula in (35), the parameter $\rho \in [0, 1]$ corresponds to the robust rank correlation coefficient known as ‘Kendall’s tau’ commonly used to summarize dependence in heavy-tailed distributions ([Nelsen, 2006](#)). If $\rho = 0$ the copula simplifies to $\mathcal{C}(u, v) = uv$ so that the ranks are independent, If $\rho \rightarrow 1$ the copula approaches $\mathcal{C}(u, v) = \min[u, v]$ so that the ranks are perfectly dependent. For simplicity we refer to ρ as the Gumbel correlation parameter.

the distribution of firm-level productivity and the operations of multi-establishment firms to match establishment-level data from the US Census of Manufactures. We calibrate the parameters governing spatial trade frictions by requiring that the model reproduce gravity regressions based on the Commodity Flow Survey (CFS).

Assigned parameters. For parsimony, we assume that the elasticity of substitution across goods within a firm equals the elasticity of substitution across firms within a given sector, $\eta = \gamma$, i.e., goods produced by different establishments under the umbrella of the same firm are just as substitutable for one another as are goods sold by other firms within the same sector. This leaves us with two elasticities of substitution. Following [Edmond, Midrigan and Xu \(2023\)](#), we set the across-sector elasticity of substitution to $\theta = 1.25$ and the within-sector elasticity to $\gamma = 10$ so that the model matches the sector-level relationship between inverse markups and sales concentration observed in US manufacturing data.⁵ This relationship can be obtained by multiplying both sides of equation (24) by market shares $\omega_{ik}(s)$ and summing across firms and locations to get

$$\frac{1}{\mu(s)} = \frac{\gamma - 1}{\gamma} - \left(\frac{1}{\theta} - \frac{1}{\gamma} \right) \text{HHI}(s) \quad (37)$$

where $\text{HHI}(s)$ denotes the sector's Herfindahl-Hirschman index of sales concentration.

Calibrated parameters. We are left with the need to calibrate the Pareto tail, the Gumbel correlation, and the trade cost coefficients for each sector

$$\xi, \rho, \delta(s)$$

We calibrate these parameters internally using the simulated method of moments. We jointly target (i) measures of national sales concentration, to pin down the Pareto tail parameter ξ , (ii) measures of the employment share of multi-establishment firms to pin down the Gumbel correlation parameter ρ , and (iii) sector-level gravity regressions to pin down the trade cost coefficients, $\delta(s)$. Importantly, we calibrate the model using measures of *national* sales concentration, not local concentration.

- (i) NATIONAL SALES CONCENTRATION. We target the average top-4 national sales shares and national sales HHI. In the US Census of Manufactures, the average top-4 national sales share for 6-digit NAICS sectors is 42% while the average national HHI is 0.10.

⁵[Edmond, Midrigan and Xu \(2023\)](#) calibrate their parameters to match the slope coefficient of this relationship, estimated in differences over time, jointly with their other parameters which target measures of concentration in 4-digit US Census of Manufactures data. For their preferred specifications, they obtain estimates of the across-sector elasticity of substitution θ between 1.15 and 1.35 and estimates of the within-sector elasticity γ between 7 and 13. We set $\theta = 1.25$ and $\gamma = 10$ as the rough midpoints of these ranges.

Table 1: Parameterization

Parameter		Value	Target
<i>Assigned values</i>			
Elas. subs. across sectors	θ	1.25	Sector-level markups and concentration (Edmond, Midrigan and Xu, 2023)
Elas. subs. within sectors	$\gamma = \eta$	10	
<i>Method of moments</i>			
Pareto tail firm productivity	ξ	10.35	National concentration
Gumbel rank correlation	ρ	0.81	Employment share of multi-estab firms
Trade cost wrt distance	$\delta(s)$		Gravity coeff. $\beta(s)$ for 3-digit NAICS

The elasticity of substitution across sectors θ and the elasticity of substitution within sectors $\gamma = \eta$ are assigned so that the model matches the sector-level relationship between inverse markups and sales concentration given in equation (37). The remaining parameters are calibrated internally by the simulated method of moments targeting the moments indicated in Table 2. The sector-specific elasticities of trade costs with respect to distance $\delta(s)$ are calibrated so that the model reproduces gravity coefficients $\beta(s)$ estimated at the 3-digit NAICS level from the Commodity Flow Survey (CFS), as in equation (38).

- (ii) OPERATIONS OF MULTI-ESTABLISHMENT FIRMS. In the US Census of Manufactures, about 4% of firms are multi-establishment and these multi-establishment firms account for about 54% of employment.
- (iii) GRAVITY REGRESSIONS. Recall that $\bar{p}_{jk}(s)\bar{y}_{jk}(s)$ denotes the value of shipments from j to k . We estimate sector-specific gravity regressions of the form

$$\ln(\bar{p}_{jk}(s)\bar{y}_{jk}(s)) = \gamma_j(s) + \gamma_k(s) + \beta(s) \ln d_{jk} + \epsilon_{jk}(s) \quad (38)$$

where $\gamma_j(s), \gamma_k(s)$ denote sector-specific source and destination fixed effects. We estimate these gravity regressions using county-to-county trade flows from the Commodity Flows Survey aggregated to the EA level for each 3-digit NAICS manufacturing sector. Our estimated slope coefficients $\beta(s)$, reported in the Appendix, measure how sensitive trade flows are to geographical distance. Goods that are more easily tradeable, such as *computers & electronics* (sector 334) and *electric equipment & appliances* (335), have estimated $\beta(s)$ that are small in magnitude. Goods that are less easily tradeable, such as *wood* (321), *petroleum, asphalt and coal* (324) and *non-metallic minerals* (327) have large negative estimated $\beta(s)$.

In the model we simulate data for each of our 363 6-digit sectors and, for each sector

Table 2: Model fit

Moments [targeted]	Data	Model
<i>National concentration</i>		
Top 4 sales share	0.42	0.44
Top 20 sales share	0.73	0.65
HHI sales	0.10	0.10
<i>Local concentration</i>		
HHI production	0.36	0.37
Top 4 sales share		0.58
Top 20 sales share		0.85
HHI sales		0.15
<i>Multi-establishment firms</i>		
Fraction multi-establishment firms	0.03	0.03
Employment share of multi-establishment firms	0.54	0.53
Sales share of multi-establishment firms	0.62	0.55

Targeted and untargeted moments in the data and their model counterparts. Moments highlighted in red are targeted by the simulated method of moments. All others are untargeted and serve as model validation. Local sales concentration has no data counterpart because it is not directly observed. The gravity coefficients $\beta(s)$ that pin down the trade cost elasticities $\delta(s)$ are reported separately in [Figure 2](#).

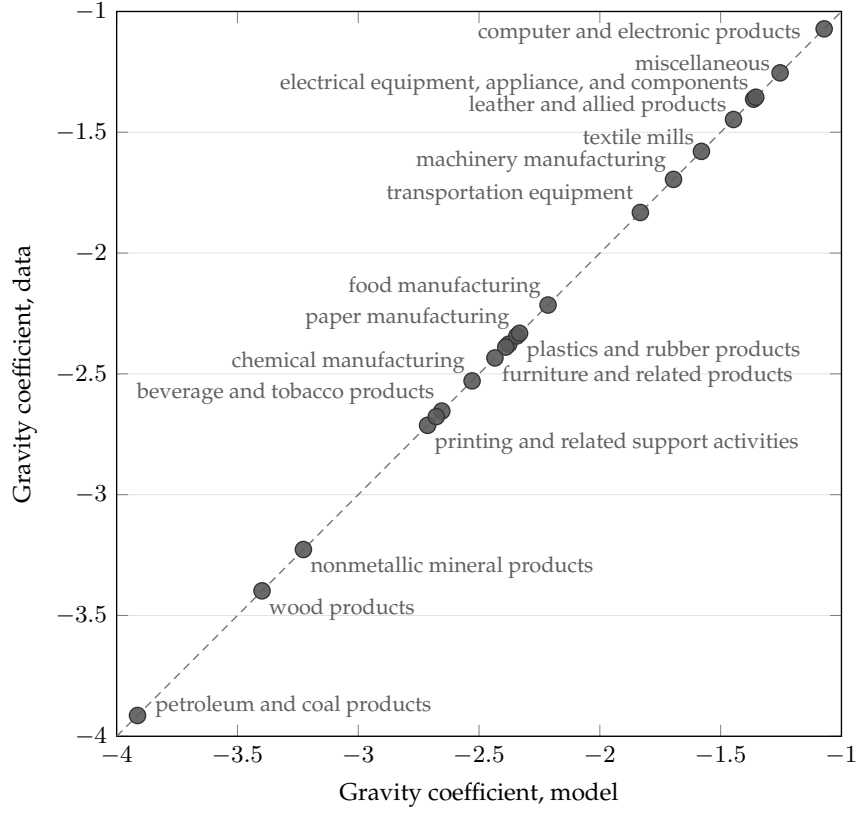
s , calculate the total value of shipments from j to k as

$$\bar{p}_{jk}(s)\bar{y}_{jk}(s) = \sum_{i=1}^{N(s)} p_{ijk}(s)y_{ijk}(s). \quad (39)$$

To be consistent with our empirical gravity regressions from the Commodity Flows Survey, we aggregate these simulated shipment flows to a 3-digit cluster of sectors and choose the parameters $\delta(s)$ in our specification (36) so that the estimated $\beta(s)$ in the model gravity regressions match their empirical counterparts from (38).

We report our internally calibrated parameters governing the productivity distribution and the operations of multi-establishment firms across locations in [Table 1](#). Jointly with our other parameters, our model matches the data on national sales concentration with a Pareto tail $\xi = 10.35$, implying considerably thinner tails than the model of oligopolistic competition in [Edmond, Midrigan and Xu \(2023\)](#), which abstracts from spatial frictions. Our model matches the 54% employment share of multi-establishment firms with a Gumbel correlation of $\rho = 0.81$ between a firm's productivity fixed effect $z_i(s)$ and its number of establishments $n_i(s)$.

Figure 2: Gravity coefficients $\beta(s)$ in model and data



Each point is one 3-digit manufacturing sector, plotting the gravity coefficient $\beta(s)$ estimated from the Commodity Flow Survey (CFS) against its model counterpart. Our model exactly reproduces the sector-level gravity effects that pin down the magnitude of spatial trade frictions. Sectors with large, negative $\beta(s)$ have *high gravity* and are less tradeable than *low gravity* sectors.

Model fit. We report key moments in the data and their model counterparts in [Table 2](#), highlighting in red the moments targeted by our calibration procedure. The model does a good job of reproducing the average amount of national sales concentration, matching the national sales HHI exactly and slightly overshooting the national top-4 sales share. The model also reproduces the employment share of multi-establishment firms almost exactly. We report the 3-digit gravity coefficients $\beta(s)$ we estimate from the CFS and their model counterparts in [Figure 2](#). Importantly, our model exactly reproduces the sector-level gravity effects that pin down the magnitude of our spatial trade frictions.

Model validation. In [Table 2](#) we also report some key moments that were not targeted in our calibration exercise. In the data, local production is much more concentrated than national sales, the local production HHI is 0.36 compared to the national sales HHI of 0.10. Our model reproduces this fact almost exactly. That said, the model undershoots the national top-20 share and the sales share of multi-establishment firms.

4 Spatial distribution of concentration and markups

In this section we present our model’s implications for the distribution of markups and sales concentration across the 170 EAs in our data. We first present our model’s implications for local *sales* concentration and show that variation in local sales concentration across locations is not well explained by variation in local *production* concentration. We then present our model’s implications for variation in markups across locations.

4.1 Local sales concentration

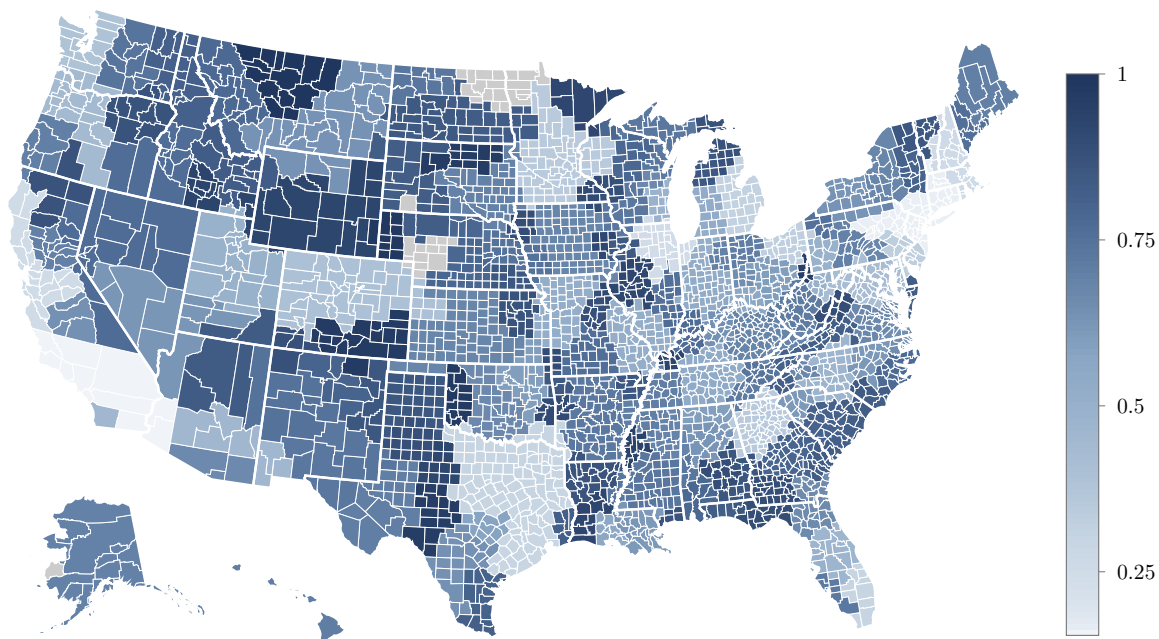
Of key interest in our framework is how much competition firms face in the destination markets that they sell to. In less competitive markets, dominant firms will be able to charge high markups. Sales concentration in local markets is not something we can directly observe with the Census data, as indicated by the blank spaces in Table 2. But given that our model does a good job of reproducing national sales concentration and matches local production concentration by construction, it seems natural to use the model to infer the amount of local sales concentration, i.e., to fill in these blanks.

Intuitively, our model implies local sales concentration lies between the national sales concentration we targeted in our calibration and local production concentration:

National Sales		Local Sales		Local Production
HHI = 0.10	<	HHI = 0.15	<	HHI = 0.37
<i>least concentrated</i>				<i>most concentrated</i>

Since the inverse of the HHI corresponds to the number of equally-sized firms, to interpret these concentration statistics more intuitively, it is as if there are on average 10 equally-sized firms nationally, about 6-7 equally-sized firms selling locally, and just under 3 equally-sized firms producing locally. The fact that local sales concentration is less than local production concentration reflects the fact that most goods are at least somewhat tradeable. While the production of goods may be quite concentrated at specific source locations, destination markets generally receive goods from a range of sources, pushing local sales concentration lower than production concentration. That said, the fact that local sales concentration is greater than national sales concentration reflects the fact that goods cannot be traded *frictionlessly*, i.e., the economy is genuinely geographically segmented. Because of this, measures of local production concentration — of the kind readily computed from data on shipments — may provide a poor guide to the amount of competition firms face in the locations where people live and consume.

Figure 3: Local production concentration (HHI)



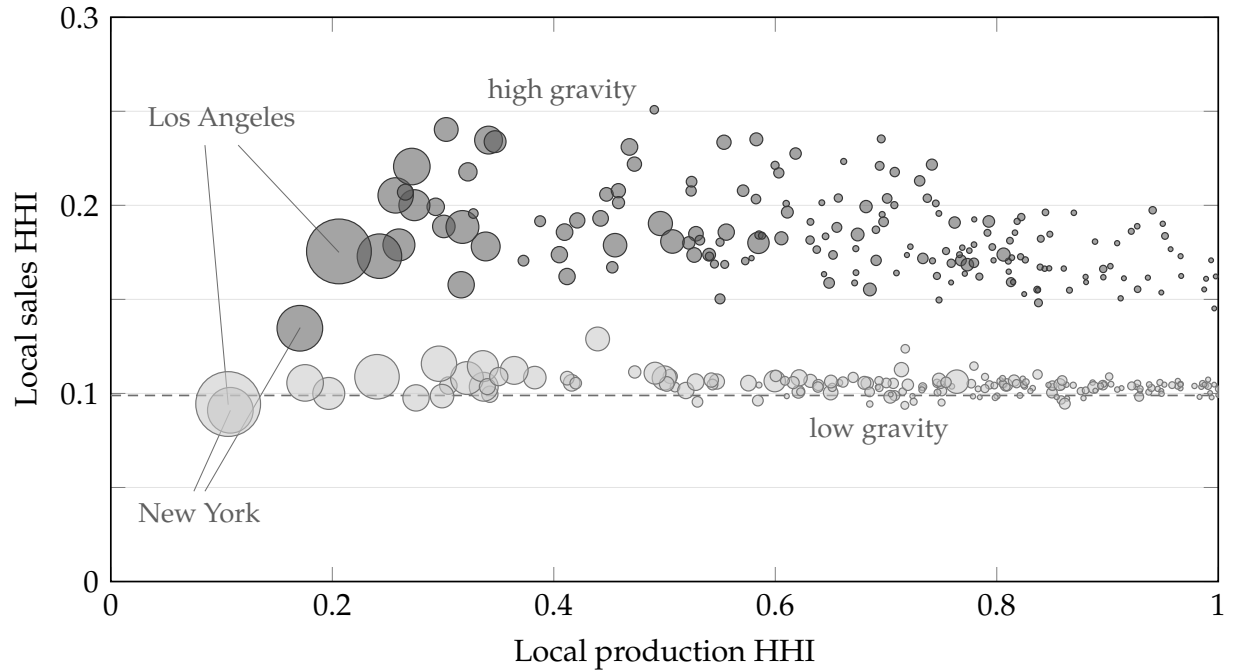
Average local production concentration, measured by the Herfindahl–Hirschman index (HHI), across the 170 Economic Areas (EAs) in our data. Production HHI ranges from about 0.16 to nearly 1, the latter indicating areas where most goods are produced by a single firm. County colors show the HHI of the EA containing each county. Counties in grey are not matched to an EA.

4.2 Sales concentration vs. production concentration

Our model predicts that local sales concentration is both lower and less dispersed than local production concentration. To assess this, [Figure 3](#) reports average local production concentration (as measured by HHIs) across the 170 EAs in our data. These range from around 0.16 to nearly 1, indicating areas where many goods are produced by a single firm. By contrast, average local sales concentration (again measured by HHIs) in our model ranges from around 0.12 to 0.17. Sales concentration in our model is indeed both much lower on average and much less variable than production concentration in the data. Moreover production concentration and sales concentration are not strongly correlated across locations. For example, greater New York has both low production and sales concentration, while the greater Seattle area has only moderate production concentration but some of the highest sales concentration in our model.

Spatial correlation of sales vs. production. [Figure 4](#) reports the scatter of observed local production concentration from our data against the local sales concentration in our model, for the 170 EAs. If local production concentration were a good predictor of local sales concentration, we would expect the scatter to cluster tightly around the 45°-line. Instead the scatter is close to horizontal. Local production concentration, of the kind we can readily

Figure 4: Local sales concentration vs. local production concentration

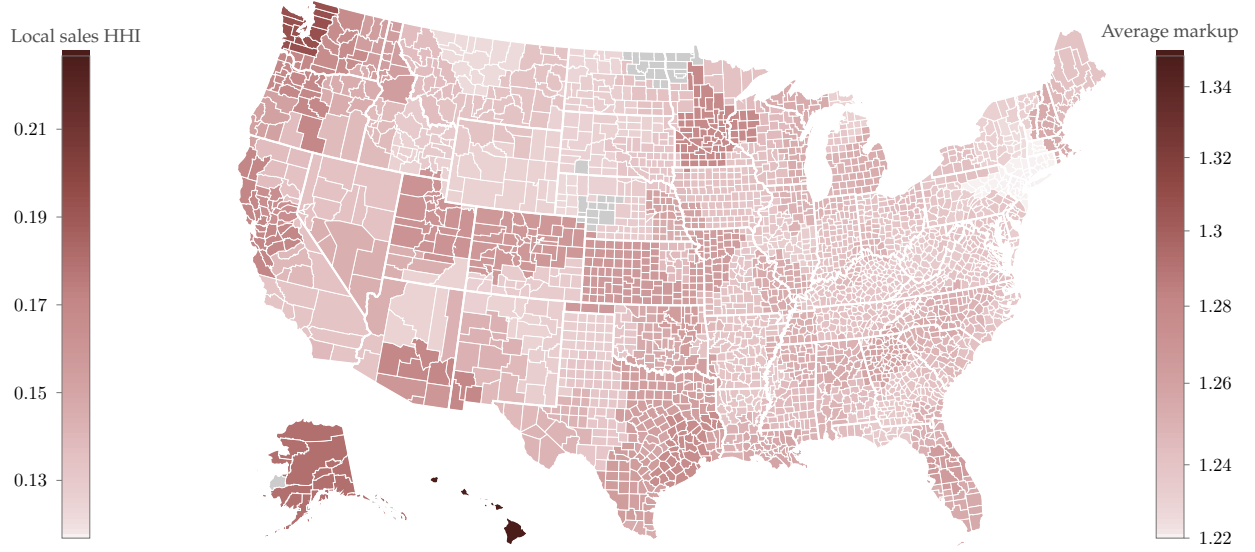


Local production HHI (from the data) against local sales HHI (from the model). Each Economic Area (EA) contributes two points, one averaging over its high-gravity (less tradeable) sectors and the other over its low-gravity (more tradeable) sectors — shown by dark and light shading respectively. Both points are then sized by the EA's total employment. If production concentration predicted sales concentration, points would cluster on the 45-degree line. Instead the scatter is close to horizontal. High-gravity points have both higher and more variable sales concentration. Low-gravity points cluster just above the national sales HHI of about 0.1.

measure from data on shipments, is simply not very informative about the local sales concentration that matters for competition and market power.

Spatial frictions and concentration. The same figure also highlights the role of tradeability. We split 3-digit sectors into high gravity sectors, with high spatial frictions that are relatively costly to ship across locations (e.g., *petroleum & coal* or *wood* products), and low gravity' sectors, with low frictions that are much less costly to ship (e.g., *computer & electronic* products), shown by the shading of each point. On average, high gravity sectors in heavier dark gray have higher local sales concentration than low gravity sectors in light gray: trade costs can substantially limit competition, especially for smaller and more geographically remote locations. High gravity sectors also have sales concentration that is considerably more variable across locations. For low gravity sectors, by contrast, local sales HHIs cluster tightly just above 0.1 — just above the national sales HHI — so that these goods are traded in what amounts to a single national market.

Figure 5: The spatial distribution of markups



Local sales concentration (sales-weighted average HHI) across the 170 Economic Areas (EAs). Because a location's average markup μ_k is in one-to-one correspondence with its average sales HHI, the same map measures both. The left axis reports the average local sales HHI and the right axis reports the corresponding average markup μ_k . County colors show the value of the EA containing each county. Counties in grey are not matched to an EA.

4.3 *Markups and sales concentration*

In our model, a location's sales-weighted harmonic average markup μ_k is in one-to-one correspondence with the location's sales-weighted arithmetic average HHI.⁶ Because of this, the spatial distribution of markups is exactly the spatial distribution of local sales concentration: a single map represents both. Figure 5 shows this map, with the left axis reporting the average local sales HHI and the right axis reporting the corresponding average markup μ_k .

Across the 170 EAs in our data, average markups μ_k range from a low of 1.22 in the greater New York area to highs of 1.29 in Anchorage and 1.35 in Honolulu. These location-specific averages μ_k mask extensive variation in markups $\mu_{ik}(s)$ across sectors s within each location k and across firms i within each sector s . Across all 6-digit NAICS sectors in our model, the lowest percentile of markups is 1.13, the median markup is 1.23, and the top percentile of markups is 1.62. The overall aggregate markup in our model is $\mu = 1.26$.

⁶Similar to (37), this follows by aggregating (24) over firms and sectors to get the location-level relationship

$$\frac{1}{\mu_k} = \frac{\gamma - 1}{\gamma} - \left(\frac{1}{\theta} - \frac{1}{\gamma} \right) \text{HHI}_k$$

between average markups μ_k and average sales concentration. Both these variables are of course endogenous, jointly determined in general equilibrium. There is no sense in which one independently causes the other.

5 Quantitative importance of spatial frictions

In this section we present our main quantitative results on the aggregate effects of spatial frictions. We quantify these effects in two complementary ways. First, we ask how much we would need to reduce the number of firms in an otherwise equivalent economy *without geography* to reproduce the level of competition we see in our benchmark. The answer is stark: it takes a 90% reduction in the number of firms to reproduce the reduction in competition created by our benchmark spatial frictions. Second, we ask how the economy changes as we move along the spectrum of spatial frictions from intranational ‘autarky’ to ‘free trade’. Our benchmark sits much closer to the free trade end of the spectrum, but even so the spatial frictions still deliver a quantitatively significant reduction in competition. We also use these results to show that our model can generate *endogenously divergent* trends in national and local concentration in response to falling trade costs, and to document the spatial distribution of markup and consumption changes in response to changes in trade costs. We show that the smallest, poorest, most remote locations generally gain more than the largest, richest, most economically central locations.

5.1 Quantifying the reduction in competition

One way to quantify the effect of spatial frictions on competition is to ask how much we would need to reduce the number of firms in an otherwise equivalent model *without geography* to reproduce the level of competition we see in our benchmark.

The ‘no geography’ model. To make this comparison, we construct a ‘no geography’ model that abstracts from spatial frictions altogether, with perfectly integrated product and labor markets, but that is calibrated to match the same national concentration facts as our benchmark. [Table 3](#) reports the markup distribution and concentration facts for this no-geography model alongside our benchmark. Markups in the no-geography model are both lower and less dispersed than in our benchmark, with the aggregate markup falling from 1.26 to 1.19. To put this in context, the aggregate markup of 1.19 in the no-geography model is even lower than the average markups our benchmark implies for New York (1.22) and Los Angeles (1.24). Abstracting from geography is like assuming the competitive conditions of New York or Los Angeles (or better) prevail *everywhere* in the country.

Changing the number of firms. We report the results of this comparison in [Figure 6](#), which shows how the aggregate markup and the productivity losses due to misallocation vary with the average number of firms per sector in the *no-geography* model. The diamonds mark an average of 600 firms per sector, as in our benchmark, and the average number of firms at which the no-geography model reproduces the level of competition in our

Table 3: Markups with and without geography

Across sectors	Benchmark model	No geography
<i>Markup distribution</i>		
p01	1.13	1.13
p10	1.15	1.14
p25	1.18	1.16
p50	1.23	1.18
p75	1.30	1.22
p90	1.41	1.29
p99	1.62	1.50
Aggregate markup	1.26	1.19
<i>Concentration (HHI)</i>		
National sales	0.10	0.10
Local sales	0.15	0.10
Local production	0.37	0.10

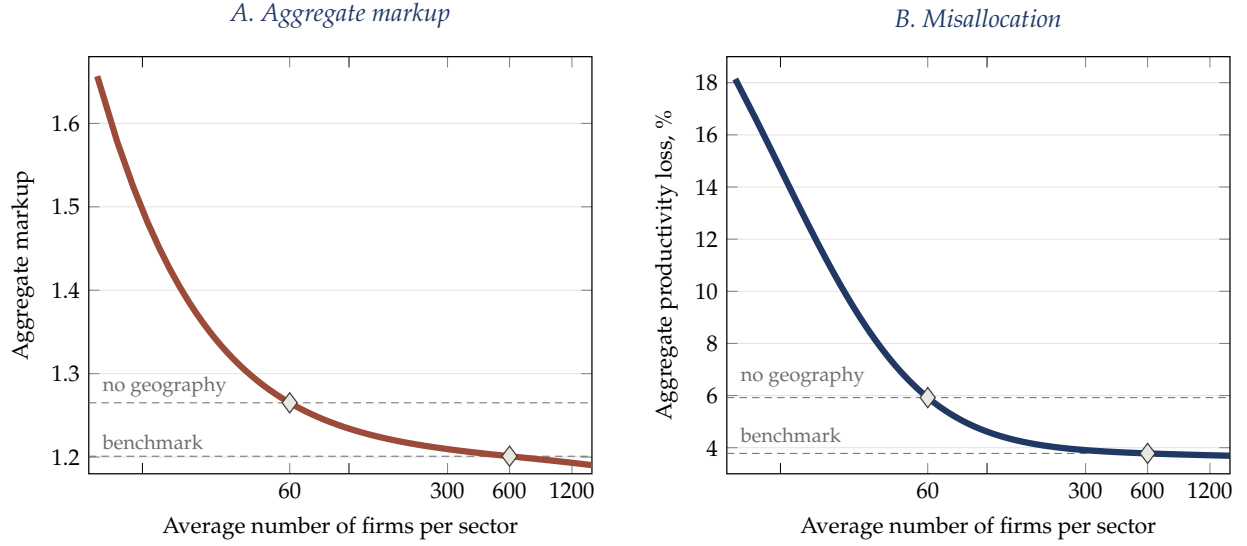
The distribution of markups across 6-digit NAICS sectors and concentration (HHI) in the benchmark model and in the no-geography model. The no-geography model abstracts from spatial frictions entirely but is calibrated to match the same national concentration facts as the benchmark. Without geography, local and national concentration coincide.

benchmark. Two features stand out. First, competitive outcomes are relatively insensitive to the number of firms over a wide range. Doubling or halving the number of firms — from 600 to 1200, or from 600 to 300 — has small effects on the aggregate markup or on the productivity losses due to misallocation. When the number of firms is already large, changes in this number have little effect on competition. Second, reducing the number of firms below about 100 raises markups and misallocation sharply. The level required to replicate the level of competition in our benchmark with geography is about 60 firms per sector — roughly a 90% reduction from the benchmark. In this sense, the reduction in competition generated by US spatial frictions is equivalent to removing a randomly chosen 90% of firms from each sector of the no-geography economy.

5.2 From autarky to free trade

A second way to quantify the effect of spatial frictions on competition is to trace out the spectrum of outcomes from ‘intranational autarky’, where trade costs are so high as to prevent any trade between locations, to ‘free trade’, where there are no trade costs at all.

Figure 6: Competition and the number of firms in the no-geography model



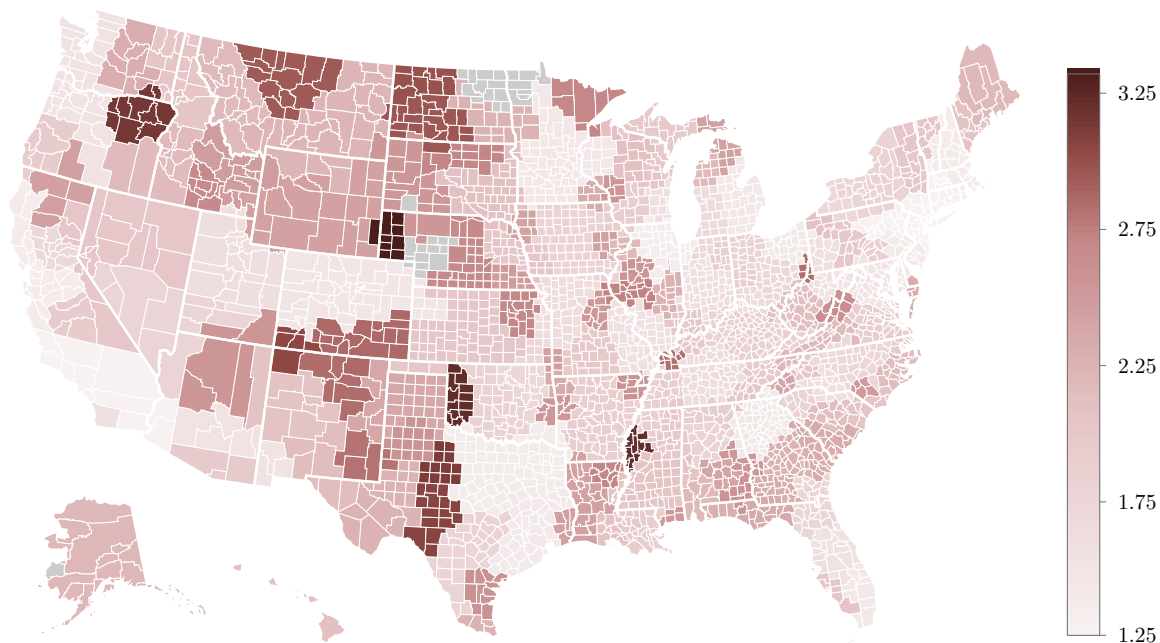
Panel A shows the aggregate markup as a function of the average number of firms per 6-digit NAICS sector in the *no-geography* model, on a log scale. Panel B shows the corresponding aggregate productivity losses due to misallocation, in percent. The marked diamonds are our benchmark calibration with an average of approximately 600 firms per sector and the much smaller count, approximately 60 firms per sector, at which the no-geography model reproduces the amount of competition in our benchmark model *with geography*.

Autarky. Under intranational autarky, with $\tau_{jk}(s) = +\infty$, markups are much larger and much more variable than in our benchmark model. Figure 7 reports our model's implications for average markups μ_k across the 170 EAs in our model for this extreme case. Without trade, local sales concentration mirrors local production concentration location by location. Smaller, less productive locations end up with much higher sales concentration and hence higher markups than in our benchmark. For example, under autarky Honolulu's average markup is 2.09 as opposed to 1.35 in our benchmark model.

The lack of trade matters less for larger, more productive locations. New York's average markup is 1.26 under autarky as opposed to 1.22 in our benchmark — New York's low markups are driven by its deep, competitive home market conditions, not import competition. A place like Seattle is a bit in between, with an average markup of 1.50 under autarky as opposed to 1.31 in our benchmark. Shutting down trade matters more for Seattle than for New York but not nearly as much as for really small and remote places.

Free trade. Under free trade, $\tau_{jk}(s) = 1$, there is no geographic segmentation in product markets. Firms in each sector compete in a single national market and set the same firm-specific markup in every destination. Average markups are therefore the same in every location, $\mu_k = \mu$, and this common level works out to be $\mu = 1.21$ — close to but not exactly equal to the 1.19 of the no-geography model, the small difference reflecting the fact that the free-trade economy still has local labor markets and differences in wages across locations. In our benchmark model large, productive locations have average markups around 1.22

Figure 7: Markups in autarky

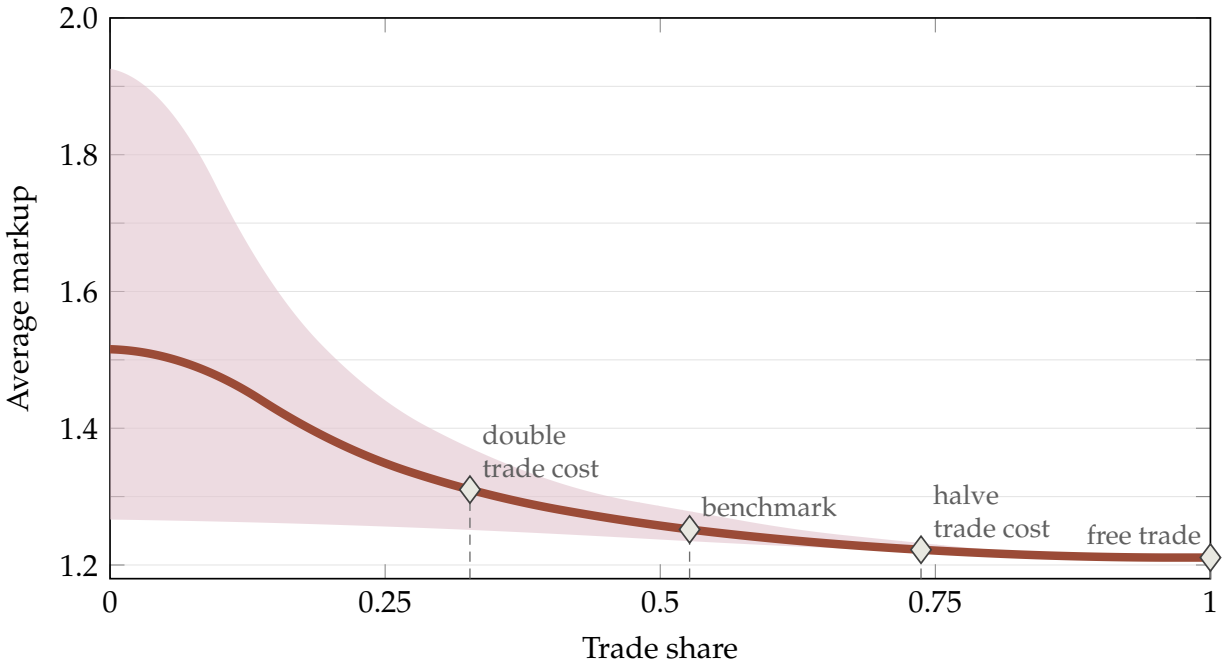


Average markup μ_k across the 170 Economic Areas (EAs) under intranational autarky, where trade costs are high enough to prevent any trade between locations. Markups are far higher and far more dispersed than in the benchmark model, ranging from 1.26 to 3.32. Without trade, local sales concentration mirrors local production concentration, so smaller and more remote locations, which lack the discipline of competition from other locations, have much higher markups. County colors show the value of the EA containing each county. Counties in grey are not matched to an EA.

(in New York) or 1.24 (Los Angeles) and so are already quite close to this free trade limit. Smaller, less productive locations are further away from this free trade limit, but overall our benchmark calibration is closer to the free trade case than the autarky case.

Between the extremes. [Figure 8](#) traces out the full spectrum between these two extremes. In this figure we report the employment-weighted average markup as a function of the employment-weighted trade share, along with the 10th and 90th percentiles of the distribution of average markups across locations. Three features stand out. First, the average markup is more responsive to increases in integration near autarky and becomes less responsive when the extent of integration is already high. Second, the spatial dispersion in markups is high when spatial frictions are high and decreases substantially as the extent of integration increases. Most of the reduction in spatial dispersion is through a compression of the upper tail of the distribution. Near autarky, the 90th percentile markups are roughly 1.5 times the 10th percentile markups. Near free trade, the entire distribution has been compressed to a single point. Third, our benchmark calibration is much closer to the free-trade end of the spectrum. But even here the remaining spatial frictions are enough to generate the quantitatively significant effects on competition, as shown in [Section 5.1](#).

Figure 8: Markups from autarky to free trade



Employment-weighted average markup as a function of the employment-weighted trade share, as spatial frictions vary from autarky (left) to free trade (right). The shaded band gives the 10th to 90th percentiles of the distribution of average markups across locations. Diamonds mark four points along the spectrum: double the trade costs, our benchmark trade costs, half the trade costs, and free trade.

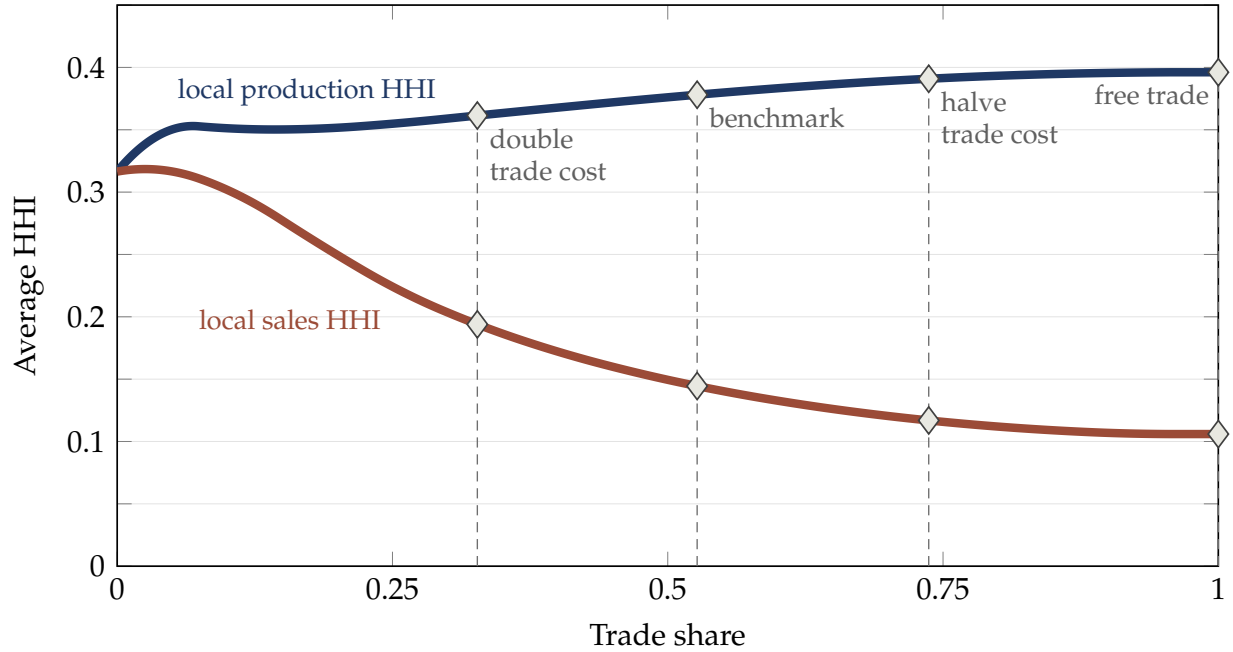
5.3 Divergence between local and national concentration

Recent empirical research has documented significantly different trends in national sales concentration and local sales concentration. National sales concentration, along with local production concentration, has been on the rise since the early 1980s. But local sales concentration has been on the decline. We now show that these divergent trends in national and local concentration emerge naturally in our model when trade costs are falling.

Concentration from autarky to free trade. Figure 9 reports the employment-weighted local sales HHI and local production HHI as a function of the employment-weighted trade share. In autarky, all goods are sold where they are produced so local sales and local production concentration coincide. As trade integration rises, local production concentration rises, as the most productive firms expand by accessing more distant markets, while local sales concentration falls, as consumers in each location gain access to a wider set of competitive suppliers. In the free trade limit, the local sales HHI converges to the national sales HHI.

Reduction in trade costs. To illustrate these effects more specifically, Table 4 reports outcomes for five points: autarky, 20% higher trade costs than our benchmark, our benchmark, half the trade costs, and free trade.

Figure 9: Concentration from autarky to free trade



Employment-weighted local production HHI and local sales HHI as functions of the employment-weighted trade share, as spatial frictions vary from autarky (left) to free trade (right). As trade integration rises, local production concentration rises while local sales concentration falls; in the free-trade limit the local sales HHI converges to the national sales HHI. Diamonds mark four points along the spectrum: double the trade costs, our benchmark trade costs, half the trade costs, and free trade.

20% lower trade costs than our benchmark, and free trade. We choose the 20% reduction to match [Coşar, Osotimehin and Popov \(2024\)](#), who find, using interregional trade data, a 15 to 20% decrease in manufacturing distance elasticities from 1963 to 2017. Moving forward from 1963-style trade costs (+20%) to our benchmark, the average national top-4 sales share rises modestly from 0.43 to 0.44 while the average local top-4 sales share falls from 0.61 to 0.58. A reduction in trade costs endogenously drives national sales concentration and local sales concentration in opposite directions. In response to lower trade costs, the most productive firms expand by accessing more distant markets. This increases national sales concentration and local production concentration. But this pattern of expansion also leads to more competition in destination markets and hence lower local sales concentration.

Divergence or convergence? It is conventional in this literature to refer to these opposite movements in national and local sales concentration as a form of *divergence* — and we have used this language too. But from the perspective of our model, the opposite-signed changes in national and local sales concentration in response to a decrease in trade costs are better understood as a kind of *convergence* towards a better-integrated national market — as trade costs fall from their benchmark levels, national sales concentration rises while local sales concentration falls. In the limit as intranational trade costs disappear, the distinction

Table 4: Effects of changes in trade costs

	Autarky	+20%	Benchmark	−20%	Free trade
<i>Markup distribution</i>					
p01	1.19	1.13	1.13	1.12	1.12
p10	1.28	1.16	1.15	1.14	1.13
p25	1.36	1.19	1.18	1.17	1.15
p50	1.48	1.24	1.23	1.21	1.19
p75	1.65	1.32	1.30	1.28	1.26
p90	1.95	1.44	1.41	1.39	1.34
p99	3.92	1.66	1.62	1.58	1.52
Aggregate markup	1.52	1.27	1.26	1.24	1.21
<i>Concentration (HHI)</i>					
National sales	–	0.10	0.10	0.10	0.11
Local sales	0.31	0.16	0.15	0.13	0.11
Local production	0.31	0.36	0.37	0.38	0.40

The distribution of markups across 6-digit NAICS sectors and concentration (HHI) for five levels of trade costs along the spectrum from autarky to free trade, including trade costs $\pm 20\%$ relative to our benchmark model. National sales concentration is undefined under autarky. National sales concentration and local sales concentration are identical under free trade.

between national and local sales concentration disappears too. This can be seen in the last column of [Table 4](#), which shows that in this free trade limit the national concentration moments exactly equal the local concentration moments.

5.4 Implications for markups and consumption

We have seen that a reduction in intranational trade costs can lead to an increase in *national* sales concentration. We now show that nonetheless this reduction makes markets more competitive and increases consumption per worker in most locations. The increase in national sales concentration is simply a byproduct of the most productive firms being able to sell in more locations. But what really matters for competitive conditions is how much competition these firms face in the markets where they actually sell their goods.

Changes in the markup distribution. [Table 4](#) reports the effects of changes in intranational trade costs on the sector-level markup distribution and the aggregate markup. Since markups are determined in large part by the amount of competition in destination markets, and local sales concentration falls as trade costs fall, it is unsurprising that reductions in trade costs lead to both lower markups and lower markup dispersion — and hence lower

productivity losses due to misallocation. Moving from 1963-style trade costs (+20%) to our benchmark, the aggregate markup falls from 1.27 to 1.26 while markup dispersion narrows. Further reductions in trade costs deliver further reductions in the aggregate markup and markup dispersion. The aggregate movements are modest — the aggregate markup falls by about 0.8% in response to a 20% reduction in trade costs from 1963 to our benchmark. But as we will now see, these modest aggregate changes mask considerable heterogeneity in markup changes across locations.

Spatial heterogeneity in markup changes. There is considerable spatial heterogeneity in how much markups respond to a given reduction in trade costs. In response to the 20% reduction in trade costs from our benchmark, the markup decreases range from around a 0.5% decrease for greater New York and around a 0.6% decrease for greater Los Angeles to around a 1.8% or 1.9% decrease in rural Utah, Colorado, and Kansas. The markup changes in more remote locations are up to around four times as large as the markup changes in more central locations. Consistent with our earlier results, the pro-competitive effects of a given reduction in trade costs are strongest for locations where competitive conditions are weakest to begin with, and more modest for locations like New York where domestic producers already compete intensely.

Spatial heterogeneity in consumption gains. The spatial heterogeneity in consumption gains is even larger. In response to a 20% reduction in trade costs, consumption per worker rises by around 1.6% in greater New York but by nearly 16% in remote parts of North and South Dakota.⁷ That is, there are ten-fold differences in consumption gains across locations, compared with four-fold differences in markup changes. The largest consumption gains are in remote locations, with more modest gains in coastal and large metro areas.

To be clear, these consumption gains from the reduction in intranational trade costs are driven by both the pro-competitive effects — with reduced markup dispersion reducing the productivity losses due to misallocation — and the standard Ricardian gains from trade that we would have in an otherwise equivalent model with constant markups. In the next section we isolate the markup channel by calculating the consumption-equivalent welfare gains from eliminating markups *holding trade costs fixed*.

6 Welfare costs of markups

In this section we quantify the welfare costs of markups in each location. We measure these welfare costs by asking how much the representative consumer in each location

⁷Since labor L_k for each location is fixed in our benchmark model, the change in final consumption C_k is equivalent to the change in consumption per worker C_k/L_k .

would gain in consumption from policies that eliminate markups. Since our benchmark model features inelastic factor supply, all of the gains from eliminating markups are due to eliminating markup *dispersion*, i.e., to reducing the productivity losses due to *misallocation*. We find that the average welfare costs of markups are large, about 5.8% in consumption-equivalent terms, and vary considerably across locations, from 1% or less in the most competitive locations to more than 20% in the least competitive locations. We then use our aggregation results to decompose these productivity losses into three distinct dimensions of misallocation: (i) *across-firms*, across firms i within source locations j , (ii) *across-sources*, across source locations j within destination k , and (iii) *across-sectors*, across sectors s within destination k . We find that the across-firm dimension is the largest, accounting for over half of total misallocation, and is remarkably stable across locations. The across-source and across-sector dimensions account for almost all the geographic variation in misallocation.

6.1 Measuring the welfare costs of markups

We measure the welfare costs of markups by asking how much the representative consumer in each location would gain from a policy that eliminates markups.

Eliminating markups. A simple policy that eliminates markups is to pay each firm i in sector s a destination-specific sales subsidy $\chi_{ik}(s) \geq 1$ to induce the firm to set establishment-level prices equal to establishment-level marginal cost. From (17) we get

$$\chi_{ik}(s) \cdot p_{ijk}(s) = \mu_{ik}(s) \cdot \frac{W_j}{z_{ij}(s)} \quad (40)$$

which evidently induces marginal-cost pricing when the subsidy equals the markup, $\chi_{ik}(s) = \mu_{ik}(s)$ for all j . The subsidy is independent of establishment location j because the firm's optimal markup is independent of establishment location. To isolate the welfare costs of markup distortions we assume that these subsidies are funded by lump-sum taxes.

Costs of markups. Table 5 reports our main results for the welfare costs of markups in our benchmark model. On average, the representative consumer would gain 5.8% in consumption-equivalent terms from eliminating markups. Since factor supply is inelastic in our benchmark model, this is entirely due to the location-specific productivity gains from the reduced misallocation that results from eliminating markup dispersion so that relative prices are aligned with relative marginal costs everywhere. When we compute the costs of markups in our model without geography, as in Section 5.1 above, we find that the representative consumer would gain 3.7% in consumption-equivalent terms from eliminating markups. Again we see that abstracting from spatial frictions leads to a quantitatively significant understatement of the welfare costs of markups. Just as

Table 5: Consumption gains from eliminating markups, %

Across Locations	Benchmark	Free trade
p01	0.9	2.0
p10	3.6	2.9
p25	3.9	3.8
p50	5.6	4.5
p75	6.9	5.7
p90	9.1	6.1
p99	14.5	7.6
Average	5.7	4.6

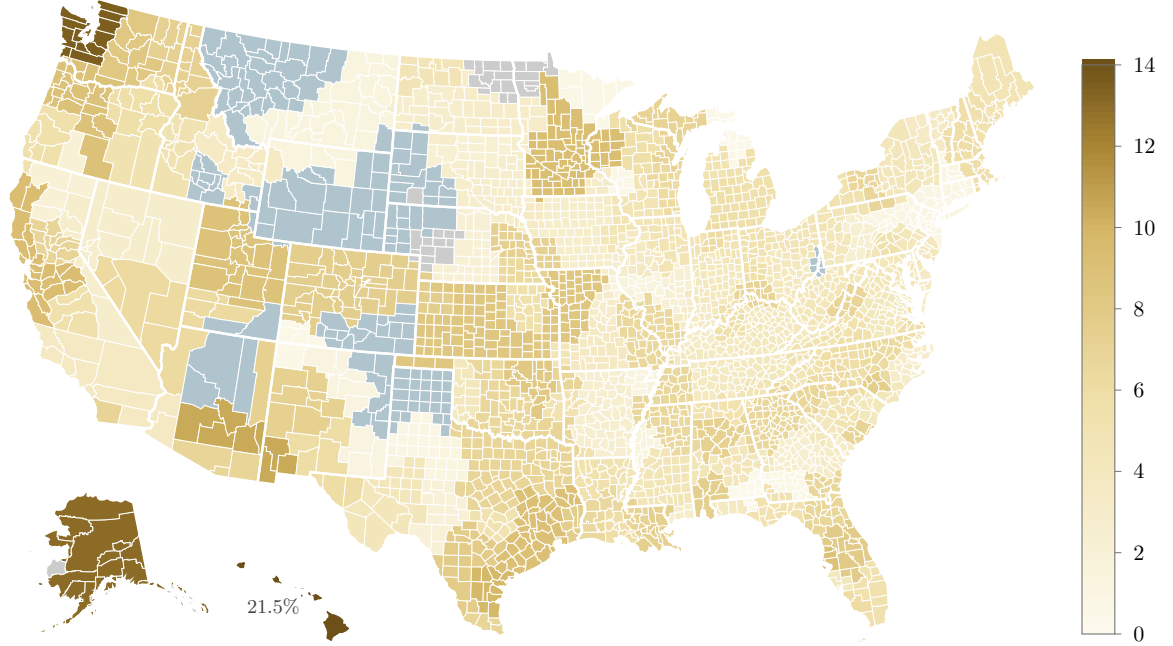
The distribution of consumption gains from eliminating markups across the 170 Economic Areas (EAs), in percent, in the benchmark model and under free trade. Gains are measured in consumption-equivalent terms. The distribution is wider in the benchmark, where spatial frictions make the welfare cost of markups much larger in some locations than others. Under free trade the gains are lower on average and less dispersed because markups are lower and less dispersed.

importantly, there is also substantial variation in the welfare costs of markups across locations — the welfare costs of markups are generally large and *very unevenly distributed*.

Spatial heterogeneity in the welfare costs of markups. To see this unevenness, [Figure 10](#) reports the full geographic variation in the consumption gains from eliminating markups across our 170 EAs. Naturally, the welfare costs of markups are larger in locations where markups are larger to begin with. Controlling for initial markups, we also find that (i) locations which have lower initial levels of consumption per worker experience larger gains, and (ii) locations which have higher initial trade shares experience larger gains from eliminating markups. In this sense, policies that eliminate markups do not just have important aggregate welfare benefits, they have stark implications for the relative gains experienced by different geographic areas, with smaller, poorer, more remote locations benefitting from the elimination of markups to a considerably greater extent than larger, richer, more central locations. Overall, we find that in the smallest, poorest, most remote locations the welfare costs of markups can be as large as 20% in consumption-equivalent terms — say 3-4 times as large as the average — but can be small, or even occasionally negative, in the largest, richest, most central locations.⁸

⁸Recall that we assume that profit income is distributed in proportion to labor income, i.e., that profit income is relatively high in high income locations. But large, highly competitive locations will have low markups. This means that when markups are eliminated, such locations experience a relatively small benefit from increased competition that can be overwhelmed by the loss of profit income resulting in a consumption loss. As can be seen from [Figure 10](#) this is a quite rare occurrence .

Figure 10: Consumption gains from eliminating markups



Percentage consumption gain from eliminating markups across the 170 Economic Areas (EAs), in consumption-equivalent terms. Gains are largest in smaller, more remote locations. In ten EAs the gain is slightly negative, ranging from 0 to -2% , and these are shown in light blue. The color scale is capped at 14% so that variation across the bulk of EAs is visible. There is one EA above this cap, Honolulu, and it is labelled with its actual value of 21.5% directly on the map. Counties in grey are not matched to an EA.

6.2 Spatial misallocation

The welfare costs of markups are driven by *productivity losses due to misallocation*, i.e., markup variation that distorts relative prices away from relative marginal costs. In our model, there is misallocation along multiple dimensions, across firms, across locations, and across sectors. We now use our aggregation results from [Section 2.2](#) above to quantify these different dimensions of misallocation.

Efficient productivity. If markup variation is eliminated then the bilateral productivity network $\bar{z}_{jk}(s)$ from equation (30) above simplifies to the familiar CES index

$$\bar{z}_j^*(s) = \left(\sum_{i=1}^{N(s)} z_{ij}(s)^{\gamma-1} \right)^{\frac{1}{\gamma-1}} \quad (41)$$

This is the *efficient* productivity level at each source j in sector s . This efficient level of productivity is independent of the destination k . Without markup distortions, the allocation across firms at a given source is the same regardless of which destinations they serve. Using these efficient levels of productivity for each j we can compute the sector-level

productivity that would prevail at each destination k if markup variation is eliminated

$$\bar{Z}_k^*(s) = \left(\sum_{j=1}^J \left(\frac{\bar{z}_j^*(s)}{\tau_{jk}(s)} \right)^{\gamma-1} \left(\frac{W_j}{\bar{W}_k(s)} \right)^{-\gamma} \right)^{\frac{1}{\gamma-1}} \quad (42)$$

Importantly, we perform this calculation *holding wages fixed* at their benchmark levels W_j . As discussed in the Appendix, we hold wages fixed to isolate the *direct* productivity effects of markup dispersion from the *indirect* effects that operate through shifts in relative wages. We then aggregate across sectors s to get a measure of the efficient level of productivity for each destination k , again keeping wages fixed at their benchmark levels

$$\bar{Z}_k^* = \left(\int_0^1 \bar{Z}_k^*(s)^{\theta-1} \left(\frac{\bar{W}_k(s)}{\bar{W}_k} \right)^{-\theta} ds \right)^{\frac{1}{\theta-1}} \quad (43)$$

Misallocation. We then measure misallocation at each destination k and each sector s by

$$\hat{Z}_k(s) := \frac{\bar{Z}_k(s)}{\bar{Z}_k^*(s)} \leq 1 \quad (44)$$

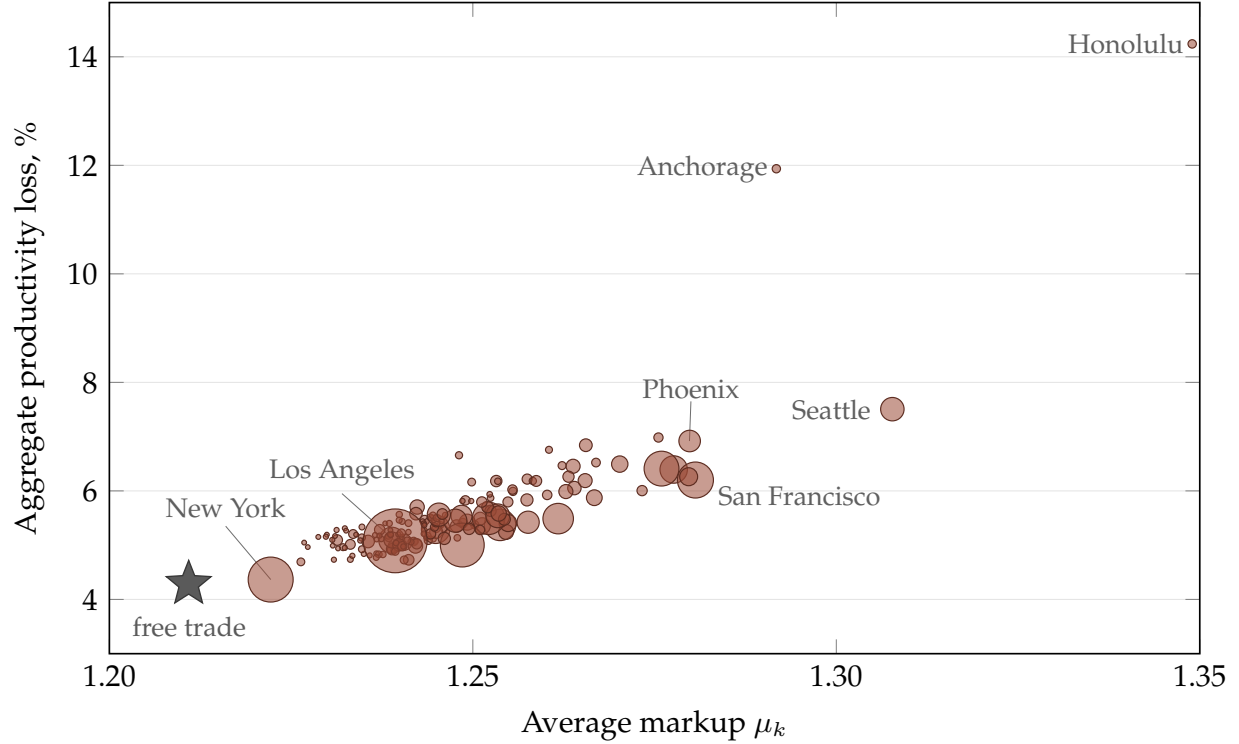
and similarly measure overall misallocation at each destination k by

$$\hat{Z}_k := \frac{\bar{Z}_k}{\bar{Z}_k^*}. \quad (45)$$

Results. Overall we find that the productivity losses due to misallocation range from 4.4% in greater New York to 14.2% in Honolulu. Locations with extensive misallocation due to markup dispersion also tend to be locations with high markup levels — the correlation between misallocation and average markups μ_k is 0.84, rising to 0.85 if we exclude the outliers Anchorage and Honolulu. [Figure 11](#) reports the full geographic variation in these losses across the 170 EAs.

Three dimensions of misallocation. Our model features three distinct dimensions of misallocation: (i) *across-firms*, across firms i within source locations j , (ii) *across-sources*, across source locations j within destination k , and (iii) *across-sectors*, across sectors s within destination k . What kind of misallocation is the most important? As shown in [Table 6](#), the across-firm component is generally the largest, averaging 3.0% across locations, and is quite stable geographically, ranging from 2.6% in Anchorage to 3.7% in Seattle. By contrast the across-source and across-sector components are generally smaller, 1.9% and 0.7% respectively, but much more variable across locations — they account for almost all

Figure 11: Productivity losses from misallocation



Aggregate productivity loss from misallocation, $1 - \hat{Z}_k$ as defined in equation (45), against the average markup μ_k , across the 170 Economic Areas (EAs). Each point is an EA, with area proportional to employment. Selected EAs are labelled. Losses range from 4.4% in greater New York to 14.2% in Honolulu and are strongly correlated with average markups. The star marks the free-trade limit, where markups and misallocation collapse to a single point.

of the cross-sectional variation in total misallocation. In a model without spatial frictions, only the across-firm dimension would be present, and average losses would be roughly 3% everywhere. The additional losses from the across-source and across-sector dimensions are due to the spatial frictions in our model and in this sense capture the *interactions* between market power and the spatial economy.

7 Labor mobility

In this section we consider an extension of our benchmark model to allow for workers to move across locations. We use a setup in the spirit of [Lagakos and Waugh \(2013\)](#) and [Kline and Moretti \(2014\)](#) where workers have different preferences for different location-specific amenities, characterized by idiosyncratic differences in Fréchet draws, to pin down labor supply to each location. One might reasonably guess that labor mobility acts to *mitigate* the welfare costs of markups — workers can now move to locations that provide higher wages and/or lower prices. But quantitatively we find that labor mobility does little to mitigate

Table 6: Decomposition of productivity losses from misallocation

	Across firms	Across sources	Across sectors	Total
p01	2.6	1.0	0.2	4.7
p10	2.7	1.6	0.3	4.9
p25	2.8	1.8	0.4	5.1
p50	2.9	1.9	0.6	5.3
p75	3.1	2.1	0.7	5.6
p90	3.3	2.3	0.9	6.2
p99	3.7	3.5	5.0	11.1
Average	3.0	1.9	0.7	5.5

The distribution of productivity losses from misallocation across the 170 Economic Areas (EAs), in percent, decomposed into three components. The across-firms component captures misallocation across firms within a source location; the across-sources component captures misallocation across source locations serving a given destination; and the across-sectors component captures misallocation across sectors within a destination. The three components sum to the total. See the Appendix for details.

the welfare costs of markups once we parameterize the model with labor mobility to match the same initial allocation of labor across locations as in our benchmark calibration.

Location-specific amenities. Each location $j = 1, \dots, J$ is characterized by an amenity value $A_j > 0$ that is common to all workers (e.g., the location’s climate). Each worker is characterized by a vector of idiosyncratic amenity draws, one for each location

$$\mathbf{v} = (v_1, v_2, \dots, v_J) \quad (46)$$

Specifically, we assume that each v_j is drawn IID from a standard Fréchet distribution

$$\text{Prob}[v_j \leq v] = \exp(-v^{-\sigma}), \quad \sigma > 0 \quad (47)$$

with tail parameter σ . As in our benchmark model, a worker that supplies labor in location j provides E_j efficiency units of labor.

Location choice. Let $c_j(\mathbf{v})$ denote the consumption of a worker of type $\mathbf{v}$ if they choose location j and let $u_j(\mathbf{v})$ denote their payoff from this choice. Consumption satisfies the individual worker’s budget constraint

$$P_j c_j(\mathbf{v}) = (1 + \bar{\pi}) W_j E_j \quad (48)$$

where as in the benchmark model we continue to assume that profit income is paid out in proportion to labor income for some constant $\bar{\pi} \geq 0$ to be determined in equilibrium. A

worker's payoff from choosing location j is then given by

$$u_j(\mathbf{v}) = A_j v_j(\mathbf{v}) c_j(\mathbf{v}) = A_j v_j(\mathbf{v}) (1 + \bar{\pi}) \frac{W_j}{P_j} E_j \quad (49)$$

where $v_j(\mathbf{v})$ denotes the specific amenity draw for location j of a worker of type $\mathbf{v}$. The problem of an individual worker is to choose a location j that maximizes their payoff

$$u(\mathbf{v}) = \max_{j=1, \dots, J} u_j(\mathbf{v}) \quad (50)$$

Labor supply. Let $\bar{L} > 0$ denote the total mass of workers. Following standard Fréchet calculations, the mass of workers supplying labor to location k is given by

$$L_k = \frac{\left(A_k E_k \frac{W_k}{P_k} \right)^\sigma}{\sum_j \left(A_j E_j \frac{W_j}{P_j} \right)^\sigma} \bar{L} \quad (51)$$

In short we have a labor supply curve, increasing in the real wage W_k/P_k for each location k , with elasticity given by the Fréchet tail parameter σ . Profit income per location cancels out because of our assumption that locations receive profit income in proportion to labor income. A higher σ reduces the dispersion in idiosyncratic amenity draws and so makes relative labor supply across locations more responsive to relative differences in real wages.

Labor market clearing. The labor market in location k clears when the total supply of efficiency units of labor $E_k L_k$ equals the total labor demand in that location

$$E_k L_k = \frac{\left(A_k E_k \frac{W_k}{P_k} \right)^\sigma}{\sum_j \left(A_j E_j \frac{W_j}{P_j} \right)^\sigma} \bar{L} = \int_0^1 \sum_{k=1}^J \sum_{i=1}^{N(s)} l_{ijk}(s) ds \quad (52)$$

Parameterization. To solve this version of the model we fix the efficiency units of labor E_k at their benchmark values and assign a labor supply elasticity of $\sigma = 2$, in line with the range of values discussed by [Fajgelbaum, Morales, Serrato and Zidar \(2019\)](#). We choose the common location-specific amenity values A_k so that the model replicates manufacturing employment L_k from the County Business Patterns aggregated to the EA level. In other words, we choose A_k so that the model with labor mobility rationalizes the allocation of employment across locations in our benchmark calibration.

Quantitative results. [Table 7](#) reports the effects of eliminating markups, across key percentiles of the distribution, for the version of the model with mobile labor and our benchmark model with immobile labor. For the model with mobile labor we report both

Table 7: Eliminating markups: labor mobility

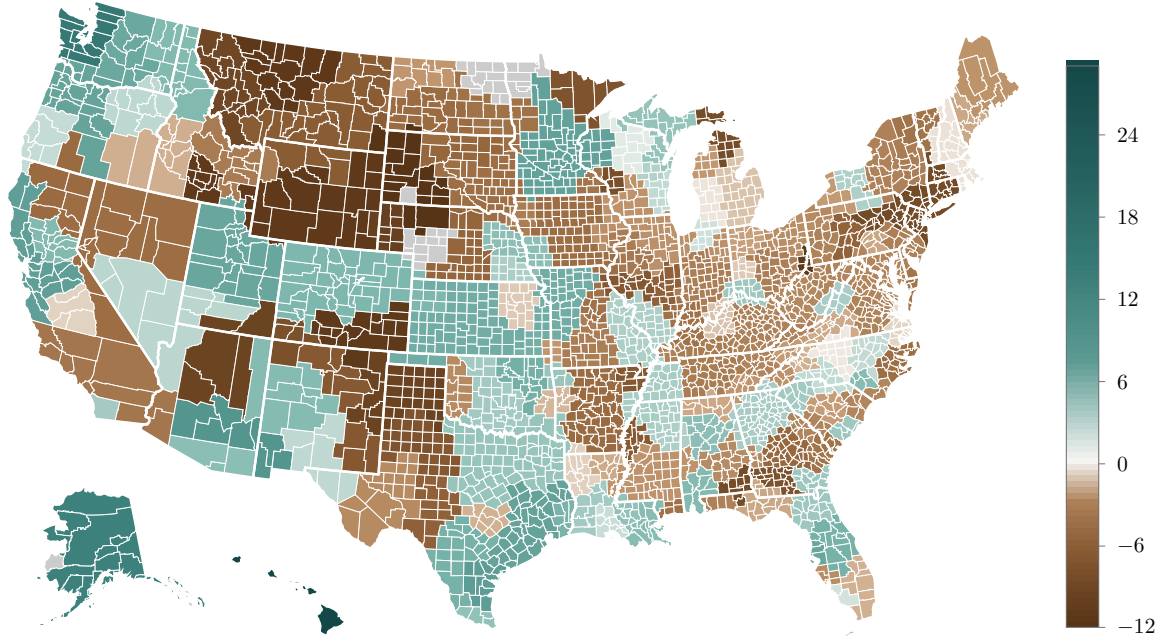
Across locations	Immobile	Mobile	
	C_k/L_k	C_k/L_k	C_k
p10	3.6	3.7	-0.4
p25	3.9	4.1	0.9
p50	5.6	5.6	5.4
p75	6.9	6.9	9.3
p90	9.1	8.9	15.6
Average	5.8	5.8	6.0

The distribution of consumption gains from eliminating markups across the 170 Economic Areas (EAs), in percent, in the benchmark model with immobile labor and in the model with labor mobility. For the model with labor mobility we report gains both in consumption per worker C_k/L_k and in aggregate consumption C_k .

the percentage changes in consumption per worker, C_k/L_k , and in aggregate consumption, C_k . In terms of consumption per worker, we find that labor mobility makes little difference quantitatively. For example, the average gain in consumption per worker from eliminating markups is 5.8% in both the model with mobile labor and the benchmark. Only in the upper tail of the distribution of consumption gains do we find noticeable differences, with the p90 gain in consumption per worker falling from 9.1% in the benchmark model to 8.9% in the model with labor mobility. In this sense, labor mobility mitigates the losses due to markups in those locations that are most severely impacted by markup distortions — but even in this upper tail the size of the mitigating effect is small. For most locations, labor mobility has even smaller effects on the welfare costs of markups.

That said, the model with labor mobility implies more substantial differences in the changes in aggregate consumption and employment across locations. For example, the p75 gain in consumption per worker is the same, 6.9%, in both models. But with mobile labor, that 6.9% gain can be decomposed into a 9.3% gain in aggregate consumption and an approximately 2% increase in employment. Likewise, the p90 gain in consumption per worker of 8.9% with mobile labor can be decomposed into a large 15.6% gain in aggregate consumption and a 6-7% increase in employment. The locations which gain the most from the elimination of markups see large increases in employment. Intuitively, there are large labor inflows to locations which are growing substantially. At the other end of the spectrum, the p10 gain of consumption per worker of 3.7% with mobile labor masks a 0.4% *decrease* in aggregate consumption and an approximately 4% decrease in employment. These locations still gain from the elimination of markups in consumption per worker terms, but they are shrinking both in absolute terms and relative to the rest of the economy. These implications for the reallocation of labor across locations are of course absent from

Figure 12: Percentage labor flows from eliminating markups



Percentage change in employment across the 170 Economic Areas (EAs) from eliminating markups, in the model with mobile labor. Locations gaining workers are shown in teal and those losing workers in brown. Aggregate labor $\bar{L}$ is fixed, so these flows redistribute workers across locations and net to zero. The locations that gain most from eliminating markups receive large inflows, drawing on labor from locations that shrink. County colors show the value of the EA containing each county. Counties in grey are not matched to an EA.

our benchmark model. [Figure 12](#) reports the full geographic variation in labor market flows across locations resulting from the elimination of markups.

8 Conclusion

We study the spatial distribution of economic activity in a quantitative model with multi-establishment firms, oligopolistic competition, and endogenously variable markups. We calibrate our model to match US Census of Manufactures firm and establishment data and intranational trade flows from the Commodity Flows Survey across 170 US Economic Areas. We show spatial frictions can have large aggregate effects, increasing both the aggregate markup and the productivity losses due to misallocation. We show that a reduction in intranational trade costs, calibrated to match long-run trends in US manufacturing, will increase national sales concentration but decrease local sales concentration. Local markets become more competitive, markups fall, and aggregate productivity rises, despite the increase in national concentration. We also show that the welfare costs of markups are large on average and very unevenly distributed. Smaller, poorer, more remote locations have costs some 20 times the costs of larger, richer, more central locations.

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Appendix

This appendix documents how we construct the datasets used to discipline trade costs, firm geography, labor endowments, and the concentration targets used in the calibration.

A. Commodity Flow Survey and Gravity Estimates

We estimate sector-specific distance elasticities based on public-use microdata from the 2017 Commodity Flow Survey (CFS). The unit of observation in the raw data is an individual shipment.

Sectors and shipment values. For each shipment we observe a detailed six-digit NAICS industry code, a shipment value in current dollars, and a survey weight. We collapse the data to three-digit NAICS industries and restrict attention to manufacturing, yielding 21 three-digit manufacturing sectors.

We form origin-destination-sector shipment totals by summing shipment values using the CFS sampling weights. Because we only use the 2017 cross section, we work in current dollars and do not deflate these flows. Origin-destination pairs with no shipments do not enter the aggregated data and are therefore absent from the regressions.

Notion of location and distances. In the gravity regressions, the notion of geography is the CFS origin and destination region. For most shipments these regions coincide with individual U.S. states. In a few cases, however, a region is defined as a multi-state metropolitan area (for example, New York-Newark, Boston-Worcester-Providence, and Washington-Arlington-Alexandria). We map each shipment to an origin region and a destination region and then aggregate weighted values by region pair and three-digit sector.

We measure distance as the great-circle distance in kilometers between region centroids. For every pair of distinct regions we observe such a bilateral distance, including pairs involving the multi-state CFS regions described above, which are treated as additional regions and handled in exactly the same way. Shipments with the same origin and destination region do not enter the gravity regressions.

Gravity specification. For each three-digit manufacturing sector s , we estimate a standard gravity equation of the form

$$\log X_{jk}(s) = \gamma_j(s) + \gamma_k(s) + \beta(s) \log d_{jk} + \varepsilon_{jk}(s),$$

where $X_{jk}(s)$ is the shipment value from origin region j to destination region k in sector s , and d_{jk} is the bilateral distance between regions j and k . The origin and destination fixed effects $\gamma_j(s)$ and $\gamma_k(s)$ absorb all location-specific determinants of trade. The slope coefficients $\hat{\beta}(s)$ discipline the distance-sensitivity of iceberg trade costs by sector in our quantitative model. We do not apply any additional trimming or winsorization to the CFS

flows beyond what is implicit in dropping origin–destination pairs with no shipments or nonpositive shipment values. [Table .8](#) reports the resulting gravity coefficients.

Table .8: Sector-level distance elasticities from CFS gravity regressions

NAICS3	Coefficient	Standard Error	Description
311	−2.215***	0.095	Food manufacturing
312	−2.654***	0.138	Beverage and tobacco product manufacturing
313	−1.579***	0.139	Textile mills
314	−2.377***	0.124	Textile product mills
315	−1.363***	0.118	Apparel manufacturing
316	−1.447***	0.133	Leather and allied product manufacturing
321	−3.398***	0.135	Wood product manufacturing
322	−2.343***	0.120	Paper manufacturing
323	−2.713***	0.116	Printing and related support activities
324	−3.914***	0.194	Petroleum and coal products manufacturing
325	−2.529***	0.109	Chemical manufacturing
326	−2.332***	0.099	Plastics and rubber products manufacturing
327	−3.227***	0.131	Nonmetallic mineral product manufacturing
331	−2.389***	0.130	Primary metal manufacturing
332	−2.677***	0.103	Fabricated metal product manufacturing
333	−1.695***	0.102	Machinery manufacturing
334	−1.072***	0.109	Computer and electronic product manufacturing
335	−1.355***	0.109	Electrical equipment, appliance, and component manufacturing
336	−1.832***	0.136	Transportation equipment manufacturing
337	−2.434***	0.107	Furniture and related product manufacturing
339	−1.254***	0.102	Miscellaneous manufacturing

B. Economic Areas, Distances, and Trade Costs

Our quantitative model is formulated at the level of BEA Economic Areas (EAs). We assign each U.S. county to a BEA EA using the BEA county-to-EA concordance based on the 1999 Economic Area definitions and aggregate all county-level information to the EA level. The Economic Areas covering the Gulf of Mexico, Puerto Rico, and the U.S. Virgin Islands are excluded, so that the dataset ultimately includes 170 EAs with positive manufacturing activity. We take as our set of model locations J the 170 EAs that receive positive manufacturing employment in County Business Patterns (CBP) and, by construction of the NETS sample below, at least one manufacturing establishment.

We measure the distance between any two Economic Areas in kilometers, using a symmetric matrix of bilateral distances. To normalize distances, we find the smallest distance observed between any two distinct EAs (79.6 km) and divide all distances by this minimum. We then also set within-EA distances to unity.

Sector-specific iceberg trade costs between EA j and EA k in sector s are parameterized as

$$\log \tau_{jk}(s) = \delta(s) \log d_{jk},$$

where d_{jk} is the normalized EA distance and $\delta(s)$ is a sector-specific distance elasticity of iceberg trade costs. We estimate a separate elasticity $\delta(s)$ for each three-digit manufacturing

sector and assign six-digit sectors in our quantitative model the value of their three-digit parent. The elasticities $\delta(s)$ are chosen so that, when we simulate the model and run analogous gravity regressions at the EA level, the implied distance coefficients in each sector match the CFS estimates $\hat{\beta}(s)$.

C. NETS Establishments and Firm-Location Structure

To discipline the geography of firms' operations we use a 2014 snapshot of the National Establishment Time Series (NETS) dataset. We begin from a national NETS extract that covers manufacturing and contains, at the establishment level, a unique establishment identifier, a unique firm headquarter identifier, the establishment's state and county FIPS code, six-digit 2014 NAICS codes, 2014 employment, and total 2014 sales. We impose the following filters in constructing our sample of manufacturing establishments:

- We drop establishments located in U.S. territories and retain only establishments located in the 50 states and the District of Columbia.
- We require a valid firm headquarter identifier and drop all establishments without a headquarter ID.
- We restrict to manufacturing establishments, defined as six-digit NAICS codes with leading digit equal to 3.
- We drop establishments with fewer than five employees, to mitigate known issues with NETS coverage and coding for very small establishments (see Barnatchez, Crane, and Decker, 2017).
- We restrict attention to firms for which the headquarter identifier corresponds to at least one observed manufacturing establishment.
- We impose a minimum size in terms of economic activity by requiring sales of at least \$1,000 in 2014.

For each remaining manufacturing establishment we retain its identifiers, location, industry code, employment, and sales. We assign each establishment to a unique BEA Economic Area using the county-to-EA mapping described above.

We define a firm as a unique headquarter identifier. Firms are assigned to six-digit NAICS sectors based on the NAICS code of their headquarters. For each firm we then construct a binary indicator for each EA that equals one if the firm has at least one manufacturing establishment in that EA and zero otherwise. If a firm operates multiple establishments in the same EA, we treat this as a single producing location in that EA. The resulting firm-by-EA incidence matrix captures each firm's geographic footprint across the United States and is the object we feed into the model as the set of locations in which each firm operates. This matrix also underlies our computation of the fraction, employment share, and sales share of multi-establishment firms. We rely on NETS primarily for its

information on locations, industry codes, headquarter structure, and employment. We use sales only in our sample restrictions and validation exercise.

After applying our filters, the 2014 NETS manufacturing sample contains 219,365 firms operating 270,141 establishments. Of these, 213,346 firms (97.2%) operate a single manufacturing establishment, while 6,019 firms (2.8%) operate more than one. Multi-establishment firms account for 56,795 establishments, or 21.0% of all manufacturing establishments in the sample. These multi-establishment firms account for 54.3% of manufacturing employment and 62.1% of manufacturing sales, so a small minority of firms accounts for a disproportionate share of activity. Manufacturing establishments are also highly concentrated across Economic Areas: the 10th, 50th, and 90th percentiles of the EA establishment distribution are 158, 582, and 2,926 establishments, respectively. The five largest EAs together host 67,344 establishments (27.8% of the total across EAs) and the largest 25 EAs host 149,247 establishments (61.6%).

D. Labor Endowments and Wages by Economic Area

Labor endowments and wages at the EA level are constructed from County Business Patterns (CBP). We treat each dataset as a cross section and use the closest available pre-2014 CBP year with consistent EA definitions. We therefore use CBP data for 2012 and aggregate manufacturing employment and the manufacturing wage bill from counties to BEA Economic Areas using the same BEA county-to-EA mapping as above. For each EA we thus obtain total manufacturing employment and total manufacturing wage payments. From the EA-level manufacturing employment and wage bill we construct:

- the employment level L_k in each EA k , given by CBP manufacturing employment, normalized so that the cross-EA mean of L_k equals one; and
- the average wage per worker in each EA, given by the CBP manufacturing wage bill divided by CBP manufacturing employment in that EA, again normalized so that the cross-EA mean equals one.

In the model, L_k is the number of workers in EA k , which we interpret as supplying $E_k L_k$ efficiency units of labor. The location-specific efficiency shifters E_k are chosen so that, at the calibrated equilibrium, the model wage bill $W_k E_k L_k$ in each EA matches the observed CBP wage bill. Beyond dropping EAs with no manufacturing activity, we do not trim or winsorize wages or employment when constructing EA-level aggregates.

Across the 170 Economic Areas, 2012 total manufacturing employment in our data is 3,469,742 workers. Employment is highly uneven: mean EA employment is 20,410 workers, with the 10th, 50th, and 90th percentiles at 426, 4,842, and 53,163 workers, respectively. The five largest EAs together account for 1,306,358 workers (37.65% of manufacturing employment), and the largest 25 EAs account for 2,594,712 workers (74.78%). By contrast, average manufacturing wages display more moderate dispersion: the employment-weighted national average wage is \$53,480 per worker, while the EA-level 10th, 50th, and 90th percentiles of the average wage distribution are \$36,500, \$47,850, and \$60,780, respectively.

E. Geographic variation in sales concentration and markups

TO BE REVISED FOR APPENDIX

If raw production concentration is not very helpful in understanding the spatial variation in sales concentration and markups, what does explain these outcomes? [Table .9](#) reports key statistics for a selection of the 170 EAs chosen to highlight the main economic forces in our model. The locations are ordered by size and span a wide range, from Los Angeles with manufacturing employment roughly 25 times the economy-wide average down to Scottsbluff at roughly 1/500th of the average. Across locations we observe two broad patterns. First, for large productive locations that have substantial domestic production and import relatively little, what matters most for local sales concentration is the structure of home supply — in particular, how many firms compete head-to-head, and how dominant the lowest-cost producer is. Second, for small unproductive locations that are forced to import almost everything, what matters most is the structure of import competition — how many outside sources are realistically cost-competitive and how concentrated import sourcing is across origins. Even so, within each broad category there is a perhaps surprisingly wide range of outcomes. New York and Seattle are both large and have similar import shares but have quite different markup levels. Honolulu and Scottsbluff are both small and import almost everything but end up with very different levels of local sales concentration and markups.

Key statistics. [Table .9](#) reports for each selected location k employment L_k , the average markup, μ_k , the average local sales HHI, the average share of expenditure on imports, the average number of plants, the average fraction of sectors with domestic production, and a number of measures of market *contestability* for each destination k .

Market contestability. Let $c_{ij}(s) = W_j/z_{ij}(s)$ denote a firm's unit cost at source j and let $C_j(s) = (\sum_i c_{ij}(s)^{1-\gamma})^{1/(1-\gamma)}$ denote the implied CES source cost index. Likewise let $C_{jk}(s) = \tau_{jk}(s)C_j(s)$ denote the trade-cost-inclusive cost index for goods delivered from source j to destination k . Then let $B_k(s)$ denote the costs of the *best* outside source for k

$$B_k(s) := \min_{j \neq k} C_{jk}(s) \quad (\text{A53})$$

and let $j_k^*(s)$ denote the source location that achieves this minimum. For each location k we then compute and report in [Table .9](#) the following summary statistics:

- **CONTEST:** The average number of outside sources j where $C_{jk}(s)$ is within 10% of home cost $C_k(s)$.
- **TOP HOME COST SHARE:** The average share of costs accounted for by the lowest cost home producer.

- **BEST GAP:** The log delivered cost advantage of the best outside source

$$\begin{aligned}\ln B_k(s) - \ln C_k(s) &= \ln C_{j^*,k}(s) - \ln C_k(s) \\ &= \ln \tau_{j^*,k}(s) + (\ln C_{j^*}(s) - \ln C_k(s))\end{aligned}$$

- **TRADE COST:** The contribution of the trade cost term $\ln \tau_{j^*,k}(s)$ to the Best Gap.
- **COMPETITIVENESS:** The contribution of the log difference in source costs $\ln C_{j^*}(s) - \ln C_k(s)$ to the Best Gap.

With these statistics in hand we review outcomes for a number of illustrative locations. First let's consider the similarities and differences between some large, productive locations.

New York. The greater New York area is extremely large, with high employment and a large number of plants covering almost every sector. In our model, New York's local sales concentration and markups are low, with an average markup of $\mu_k = 1.22$ and a sales HHI of 0.12. The import share is relatively low, 0.36. On average the best outside source of supply has delivered costs 0.23 log points higher than the costs of domestic supply, and on average the best outside source of supply is only slightly, -0.04 log points, more competitive than home supply. Cost-realistic outside options are rare: the average number of other locations that have delivered costs within 10% of domestic costs is 0.284, i.e., there is generally less than 1 effective outside competitor location. The low import share therefore reflects the difficulty of penetrating the New York market given the combination of sizeable trade costs and low domestic costs. But despite the lack of import discipline, concentration stays low because home supply features extensive head-to-head competition — there are roughly 61 plants per sector on average, and the lowest-cost home plant accounts for 'only' about 51% of home supply in cost terms. Overall, greater New York's low local sales concentration reflects the strong competition amongst its many domestic producers, not import contestability.

But not all large productive locations have strong competitive home conditions.

Seattle. The greater Seattle area is also large and productive with home production in over 90% of sectors. Like New York, imports provide little discipline, the best outside source has delivered costs 0.38 log points above home costs and the number of cost-realistic outside locations is just 0.41. But unlike New York, in our model Seattle has high markups and sales concentration, with an average markup of $\mu_k = 1.31$ and a sales HHI of 0.19, the highest we see among the large locations. What makes Seattle different is the structure of its home supply. The lowest-cost home plant accounts for considerably more of home supply, about 66% in cost terms compared to 51% for the lowest-cost home producer in New York, and there are only about 10 plants per sector compared to about 61 in New York. With fewer plants and a more dominant low-cost producer in each sector, there is less head-to-head competition among domestic firms, which translates directly into

higher concentration and higher markups. In short, Seattle shows that being large and productive is not sufficient for low concentration, what really matters is the depth and competitiveness of home supply.

What about some smaller, less productive locations? Are they fated to have weak competitive conditions?

Scottsbluff. Scottsbluff is a tiny location on the Nebraska/Wyoming border. Home production is minimal, with fewer than one plant per sector on average and home production present in only 13% of sectors. As a result, Scottsbluff is a near-complete importer with an import share of essentially 1.00. One might expect such dependence on imports to leave Scottsbluff vulnerable to high markups, but in fact the opposite is true — the average markup is $\mu_k = 1.23$ and the sales HHI is 0.12, both among the lowest we find. This happens because Scottsbluff's imports are intensely contested. Even though Scottsbluff is in the middle of the country, it happens to be roughly equidistant from many production centers. The number of cost-realistic outside locations is nearly 48, one of the largest we see. With so many suppliers competing head-to-head, no single firm dominates. That said, while Scottsbluff has strong competitive conditions, the low markups do not mean that Scottsbluff is cheap. The CES price index works out to be $P_k = 0.92$, well above locations like New York ($P_k = 0.70$) and Minneapolis ($P_k = 0.76$) which have similar average markups, reflecting the high cost of delivering goods to a remote location with almost no domestic production.

But sometimes being small and remote really just means what you expect.

Honolulu. Honolulu is a mid-sized location that, like Scottsbluff, has limited home production and is heavily reliant on imports, with an import share of 0.62. But where Scottsbluff has low sales concentration because of its diversified and highly-contested imports, Honolulu has the opposite: in our model, Honolulu's local sales HHI is 0.23 and the average markup is $\mu_k = 1.35$, among the highest we find. Scottsbluff is remote, but not overwhelmingly so, and more importantly is comparably close to many competing sources of supply. Honolulu is *much* more remote. Honolulu's trade costs are much larger than elsewhere — the trade cost component of the best outsider gap is 0.98, roughly double that of any mainland location. Even though the best outside source is substantially more cost-competitive than home in net terms, about -0.25 log points, the trade costs overwhelm this advantage, leaving the best outsider 0.74 log points more expensive than home overall. Because of this, cost-realistic outside options are rare (there are only about 0.92 such locations) and home supply, where it exists, is concentrated (the lowest-cost home plant accounts for 79% of home supply in cost terms). The combination of remoteness, low contestability, and incomplete home production coverage allows a few suppliers to dominate, sustaining the highest markups and concentration of any location in our model.

These four examples illustrate the key forces that operate, in various combinations, across all 170 locations in our model. Locations with low contestability and concentrated

home supply tend to have higher markups, while locations with either deep home competition or diversified import sourcing tend to have lower markups — though as the contrast between New York and Seattle or Scottsbluff and Honolulu makes clear, the finer details of each location's geography and production structure still play an important role in determining their individual location-level outcomes.

But how much does all this matter in the aggregate?

Table .9: Characteristics of Select Locations

Location	L_k	μ_k	HHI	Import Share	Plants	Coverage	Contest	Top 1 Share	Best Gap	
									Trade Cost	Comp.
Los Angeles	24.9	1.24	0.13	0.22	56.79	0.98	0.223	0.505	0.277	0.043
New York	12.3	1.22	0.12	0.36	61.02	0.98	0.284	0.511	0.270	-0.044
San Francisco	7.84	1.28	0.17	0.26	24.98	0.95	0.143	0.598	0.306	0.084
Minneapolis	4.49	1.28	0.17	0.39	14.71	0.93	0.597	0.650	0.473	-0.091
Seattle	3.24	1.31	0.19	0.37	10.21	0.91	0.409	0.657	0.426	-0.049
Phoenix	2.71	1.28	0.17	0.50	8.61	0.86	1.109	0.680	0.372	-0.107
San Antonio	0.641	1.26	0.16	0.77	3.85	0.76	5.107	0.766	0.336	-0.342
Honolulu	0.298	1.35	0.23	0.62	1.74	0.55	0.917	0.786	0.983	-0.249
Anchorage	0.289	1.29	0.18	0.77	0.99	0.36	0.339	0.782	0.906	-0.240
Redding	0.027	1.24	0.14	0.97	0.52	0.27	17.43	0.856	0.367	-0.642
Minot	0.010	1.23	0.12	0.97	0.24	0.15	20.07	0.897	0.498	-0.560
Scottsbluff	0.002	1.23	0.12	1.00	0.18	0.13	47.63	0.935	0.442	-0.916

Notes: Locations are ordered by L_k (manufacturing employment relative to the economy-wide average). Markup is the sales-weighted average markup μ_k . Sales HHI is the sales-weighted average local Herfindahl index. Import Share is the sales-weighted average import penetration. Plants is the average number of production plants per sector. Home Coverage is the expenditure-weighted fraction of sectors with home production. Contest. is the number of outside origins within 10% of home delivered cost. Top 1 Home Cost Share is the cost share of the lowest-cost home plant. Trade Cost and Competitiveness are the two components of the log delivered cost advantage of the best outsider relative to home: Trade Cost is $\log \tau_{j^*k}$ and Competitiveness is $\log C_{j^*} - \log C_k$, so that the best outsider gap $\log B_k - \log C_k$ equals Trade Cost + Competitiveness. All statistics are sales-weighted averages across sectors. Location names correspond to BEA Economic Areas: Los Angeles = Los Angeles-Riverside-Orange County, CA-AZ; New York = New York-North New Jersey-Long Island, NY-NJ-CT-PA-MA-VT; San Francisco = San Francisco-Oakland-San Jose, CA; Minneapolis = Minneapolis-St. Paul, MN-WI-IA; Seattle = Seattle-Tacoma-Bremerton, WA; Phoenix = Phoenix-Mesa, AZ-NM; San Antonio = San Antonio, TX; Honolulu = Honolulu, HI; Anchorage = Anchorage, AK; Redding = Redding, CA-OR; Minot = Minot, ND; Scottsbluff = Scottsbluff, NE-WY.

Table .10: Decomposition of Productivity Losses: Selected Locations

Location	Across-Firms	Across-Sources	Across-Sectors	Total
Los Angeles	3.4	0.9	0.8	5.1
New York	3.1	0.8	0.5	4.4
San Francisco	3.8	1.7	0.8	6.2
Minneapolis	3.5	2.0	1.0	6.4
Seattle	3.7	2.4	1.6	7.5
Phoenix	3.5	2.3	1.2	6.9
San Antonio	3.2	2.3	0.8	6.3
Honolulu	3.0	4.4	6.9	14.2
Anchorage	2.6	3.7	5.8	11.9
Redding	2.9	1.8	0.7	5.4
Minot	2.6	2.0	0.6	5.2
Scottsbluff	2.7	1.7	0.6	5.0

Figure A13: Sector-Level Misallocation: New York vs. Honolulu

